



3 INTRODUCTION [1]

3.1 AIRCRAFT

An aircraft is machine or vehicle that can fly through air and is powered by engines.

3.2 TYPE OF AIRCRAFT

Aircraft are categorized into two different types such as

3.2.1 Manned Aerial Vehicle

These are aircraft with pilots or crew members on board to control and navigate the flight.

They include a wide range of aircraft types such as

3.2.1.1 Fixed Wing Aircraft

(Commercial Aircraft, Cargo Aircraft, Private Aircraft, Gliders).

3.2.1.2 Rotary Wing Aircraft

(Helicopter).

3.2.1.3 VTOL Aircraft

(Vertical Takeoff and Landing).

3.2.1.4 Spacecraft

(Space Shuttle).

3.2.2 Unmanned Aerial Vehicle

These are remotely controlled or autonomous aircraft that do not carry pilots onboard.

They include a wide range of aircraft types such as

3.2.2.1 Military UAVs

(Design for Combat such as MQ-9 Reaper).



3.2.2.2 Civilian UAVs

(Design for Aerial photography and Delivery services).

3.3 BASIC COMPONENTS OF AIRCRAFT

Basic components of aircraft are given below

- 1) Wing.
- 2) Fuselage.
- 3) Empennage.
- 4) Engine.
- 5) Landing Gear.

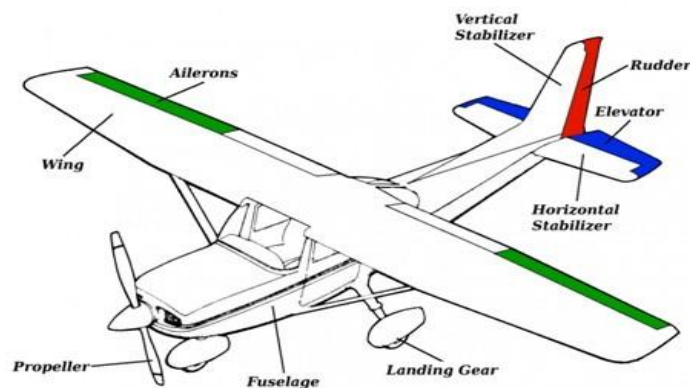


Figure 3-1 Basic Aircraft Components

3.3.1 Wing

The purpose of wings on an aircraft is to generate lift. It can generate a lift on the principle of Bernoulli's Principle. According to Bernoulli's Principle, "Pressure and Velocity is inversely proportional to each other". When air moves faster, the pressure becomes lower. At upper surface of aircraft speed of airflow is increasing as compared to speed of airflow at the bottom surface of aircraft. Pressure at the bottom is higher than the pressure at the top. Due to this difference in pressure upward force (Lift) is generated on aircraft.

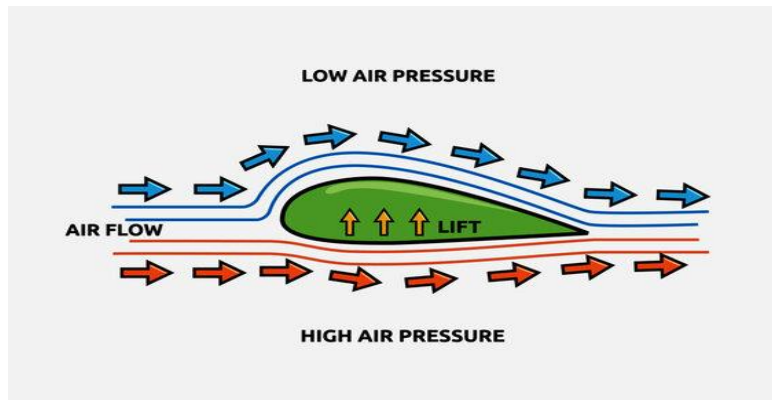


Figure 3-2 Bernoulli Principle Flow Diagram

3.3.1.1 Structure of Wing

The wing structure of an aircraft is designed to create a lift while providing strength and stability. Structural components of wing of aircraft are given below

3.3.1.1.1 Spar

The main structural beam runs along the length of the wing, providing strength and holding the wing's shape.

3.3.1.1.2 Rib

Rib can support structures that run perpendicular to the spar, shaping the wing and distributing loads.

3.3.1.1.3 Skin

The outer covering provides a smooth surface for airflow and additional strength.

3.3.1.1.4 Stiffeners

Stiffeners are important for maintaining the structural integrity of an aircraft and prevent skin from buckling while keeping it lightweight.

3.3.1.1.5 Ailerons

Ailerons are movable surfaces on the trailing edge of the wing, ailerons control roll during flight.

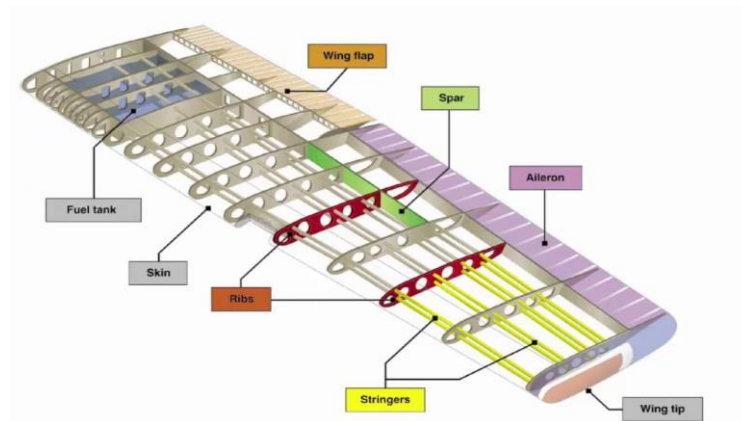


Figure 3-3 Wing Internal Structure

3.3.2 Fuselage

The fuselage is the main body of an aircraft. Fuselage provides the primary structure that supports the wings, tail and distributing loads and stresses during flight. The shape of the fuselage is designed to minimize drag, allowing the aircraft to move smoothly through the air.

3.3.2.1 RC Plane Fuselage

The fuselage of a remote-controlled (RC) plane serves similar purposes, it also provides support for wing, tail and other components. Remote Control (RC) Aircraft fuselage contains various essential components, including

Radio Gear (Receivers and servos that control the plane movements.)

Battery (Power source for the electric motor or servos.)

Motor (In electric RC planes, the motor is mounted in the front section of the fuselage.)

3.3.3 Empennage

The empennage of an RC plane refers to the Tail section such as Horizontal Stabilizer, Vertical Stabilizer, and Control surfaces like the Elevators and Rudder. The empennage helps to maintain the aircraft stability during flight. The horizontal stabilizer counteracts



pitch movements (nose up or down), while the vertical stabilizer counteracts yaw movements (side-to-side).

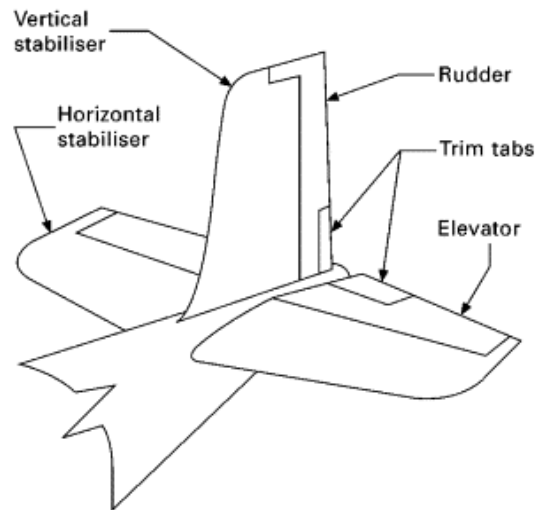


Figure 3-4 Empennage Diagram

3.3.4 Landing Gear

The landing gear provides the necessary support for the plane during takeoff and landing phases, allowing it to operate safely on the ground. Landing gear (Tail-Dragger configurations) can make flying of the RC plane easy, making it more user-friendly for pilots.

3.3.5 Engine

The engine in an RC (remote-controlled) plane is a critical component, it provides thrust, enabling the aircraft to move forward and achieve flight. Engine allows the pilot to control the speed of the aircraft by adjusting the throttle, the pilot can increase or decrease the thrust. We can use Electric Motor as an engine in RC planes because of their various sizes and power rating, suitable for different types of models.



How to control the movement of an Aircraft?

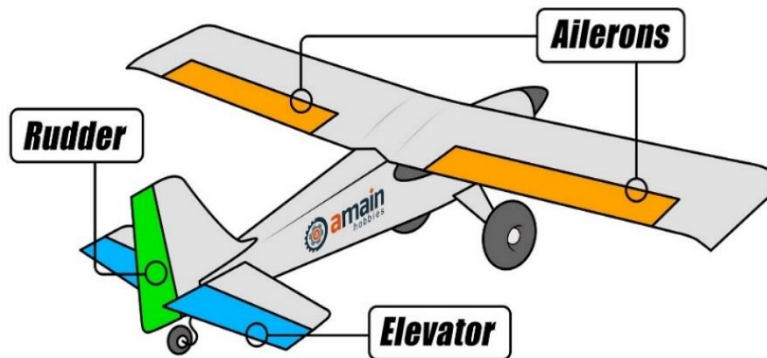


Figure 3-5 Control Surface Visualization

Controlling the movement of an RC (remote-controlled) plane involves using a **transmitter** (the remote control) and various control surfaces on the aircraft. Basic control of an RC plane is given below

3.3.6 Basic Controls of an RC Plane

3.3.6.1 Throttle Control

It controls the engine power and is typically controlled by a stick or slider on the transmitter. Increasing throttle makes the plane go faster and climb; while decreasing it slows the plane down and can cause it to descend.

3.3.6.2 Ailerons

Ailerons control the roll of the aircraft, allowing it to tilt left or right. When the right stick on the transmitter is moved to the left or right, one aileron goes up while the other goes down, causing the plane to roll in the desired direction.

3.3.6.3 Elevators

Elevators control the pitch of the aircraft, allowing it to climb or descend. Pulling the right stick back raises the elevators, causing the nose to tilt up (climbing), while pushing it forward lowers the elevators, making the nose tilt down (descending).



3.3.6.4 Rudder

Rudder controls the yaw of the aircraft, allowing it to turn left or right. Moving the left stick (or a separate rudder control) to the left or right deflects the rudder, causing the plane to turn in the desired direction.

3.3.6.5 Trims

Trim controls are used to make fine adjustments to each control surface (elevators, ailerons, rudder) to maintain straight and level flight without constant input.



4 CONCEPTUAL DESIGN [2]

During the conceptual design phase, the team identified key design objectives by analyzing the scoring equations and competition requirements. The highest-scoring aircraft configuration was then derived by comparing relevant Figures of Merit (FoM). The final conceptual design is a high-wing monoplane, fuselage mounted tractor, with a conventional tail and tricycle landing gear, as presented in Section 3.5.

4.1 MISSION REQUIREMENTS

The team thoroughly analyzed the competition criteria and the Student Aircraft System (SAS) to determine an optimal configuration with the highest potential to maximize the overall score while adhering to design constraints. Additionally, a sensitivity analysis of the scoring system was performed to identify the key factors that have the most significant impact on the total score.

4.1.1 Mission 1: Without Payload

We will make sure that the aircraft takes off within the designated 50 ft runway and completes the required maneuvers: two circuit patterns, one aileron roll in each direction (clockwise and anticlockwise), and an inside loop. We will aim to finish these maneuvers within the 5-minute time limit, knowing we will have a maximum of three attempts, with each retry resulting in a marks deduction. The mission clock will start as soon as we advance the throttle for the first take-off attempt. We understand that performance will be normalized across all teams who successfully complete the mission, and we will aim for a smooth landing in the prescribed area to earn additional points.

4.1.2 Mission 2: With Internal Payload

We will ensure that the payload for the mission weighs exactly 200 grams, giving us the freedom to choose its dimensions and material, if it remains a separate entity from the



aircraft's structural and functional components. The aircraft will take off within a 50 ft runway while carrying the securely fastened payload to prevent any movement during flight. Our required maneuvers will include two complete circuit patterns, with each circuit covering approximately 100 meters above the main field. During the second circuit, the aircraft will perform an inverted fly-pass through the field and fly through the designated Flight Box before preparing for a final approach and landing. We will complete all maneuvers within a 5-minute time limit.

4.1.3 Mission 3: With Internal Payload Dropping

We will aim to drop at least one payload into the designated target area on the airfield, using the Payload Dropping Zone diagram for clarity. There will be no limit on the number of payloads we can drop; however, we will ensure that the total payload weight adheres to the aircraft's maximum weight constraint. Each payload will be a rigid, separate entity placed within the aircraft, with dimensions exceeding 2x2x2 inches and a maximum weight of 200 grams. We can choose to drop the payloads with or without a parachute, knowing that extra credit will be awarded for parachute performance based on design, dynamics, and effectiveness. The drops will occur from an altitude of at least 20 feet, and we will release only one payload at a time, requiring the aircraft to maneuver back for each drop. We will complete the mission within 7 minutes, and a successful landing in the designated area will earn us additional points.

4.2 PAYLOAD REQUIREMENTS

The payload design for the aircraft must adhere to specific guidelines to ensure compliance and performance efficiency. Each payload must be a rigid, separate entity that fits within the aircraft and can be released individually. Although there's no limit on the number of payloads, the combined weight must respect the aircraft's maximum weight capacity, and each payload cannot exceed 200 grams.

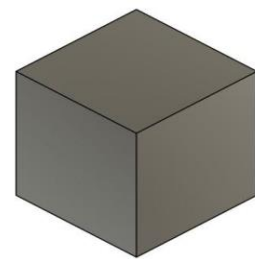


Figure 4-1 Payload dimension



Additionally, the minimum dimensions for each payload are 2x2x2 inches in any direction, allowing teams flexibility in shape and material selection, if the payload remains within these constraints. A carefully engineered release mechanism is essential to drop one payload at a time, ensuring stability and control during deployment. By considering lightweight and durable materials for payload construction, teams can maximize payload capacity while optimizing overall performance.

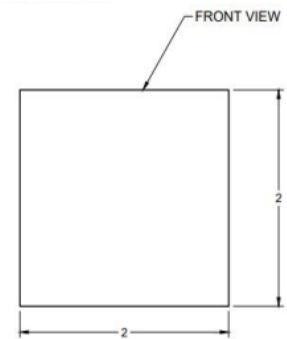


Figure 4-2 Front View of Payload

Both the front view and top view confirm that each side measures exactly 2 inches, ensuring that the payload meets the minimum dimensional constraints. This size specification allows flexibility in material selection while maintaining a standard shape for consistent fit and placement within the aircraft. The cubic shape provides a stable structure for secure storage and deployment, aligning with the project's payload requirements.

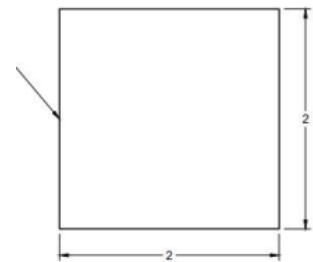


Figure 4-3 Top View of Payload

4.3 DESIGN CONSIDERATIONS

Our aircraft is carefully designed to meet the specific challenges of the three DBFC missions, prioritizing agility, payload capacity, and maneuver precision. Key features include lightweight construction for an ideal thrust-to-weight ratio, a dependable payload drop mechanism for Mission 3, and a focus on aerodynamic efficiency and stability to ensure smooth flight and precise landings.

4.4 DESIGN CONSTRAINT

Design Constraints are given below on the table.



Table 4-1 Design Constraints

Category	Requirement
Configuration	Wingspan cannot exceed 6.5 ft (2.0 m).
	The aircraft can be designed in any configuration
	Weight must not exceed 3 kg (including payload(s) for all missions).
Payload	The Team can choose the dimensions and material of the design.
	The payload must weigh exactly 200 grams.
	The payload must be a separate entity, not a part of the structural or functional components.
Propulsion	Aircraft must be propeller driven and electric powered
	Maximum battery capacity: Rated at 5000 mAh or less.
	Only Lithium Polymer (Li-Po) batteries are allowed.
	Voltage: Rated at 11.1 V (3S) or less.

4.5 AIRCRAFT CONFIGURATION

An optimal aircraft configuration and its components were determined through the down select process outlined in Table below. Selected parameters are listed below along with their impact on scoring.

Table 4-2 Parameter Detail for Selection

Step 1	Identify high-level aircraft subsystems that require a configuration down select.
Step 2	Select relevant parameters for each subsystem and assign a weightage for importance.
Step 3	Grade configuration options with a 0-5 grade in comparison to baseline configuration.
Step 4	Calculate total score for each option by multiplying weights with grades and summing the scores.



4.5.1 Monoplane

A monoplane is an aircraft with one set of wings attached to the body. Its main advantages are that it's easier to build, has a good lift, and is simpler to control. Because it's lighter, it's also more efficient than other designs.

4.5.2 Biplane

A biplane features two wings stacked vertically with a gap between them to allow airflow. While this configuration generates more lift than a monoplane, it also produces higher drag.

4.5.3 Flying Plane

A flying plane is a fixed wing tailless aircraft. It has no definite fuselage and while it had a promising L/D ratio, it constricted us in many areas of performing missions.

4.5.4 Selected Configuration

Table 4-3 Wing Configuration

				
Parameter	Weight (%)	Monoplane	Biplane	Tandem
Stability	30	5	4	3
Drag	30	4	3	2
Ground Clearance	10	5	4	3
Maneuverability	20	3	4	4
Manufacturability	10	5	3	2
		4.3	3.6	2.8



Monoplane designs are pivotal in the evolution of aviation. Their aerodynamic advantages, structural simplicity, and versatile performance make them the preferred choice for modern aircraft.

4.6 PROPELLER CONFIGURATION

4.6.1 Fuselage Mounted Pusher

In a pusher configuration, the propeller is at the back, giving smoother airflow over the aircraft, which makes it more efficient and good for long missions without carrying cargo. The back propeller also keeps the area under the aircraft clear for dropping items. But this design can make it harder to cool the motor, especially on long or powerful flights, and can affect stability and control during tricks like loops, where precise thrust is important.

4.6.2 Wing Mounted Twin Tractor

The wing-mounted twin tractor configuration, with two propellers on either side of the wing, enhances maneuverability and control authority due to additional thrust and distributed power. This setup provides redundancy, allowing continued flight even if one motor fails, which is beneficial for payload missions requiring high reliability. Forward-mounted propellers improve cooling and enable efficient control, particularly in maneuvers like inside loops. However, this configuration adds weight and structural complexity, potentially leading to mechanical issues. The distributed propulsion also increases drag, lowering efficiency during extended flights without payload.

4.6.3 Fuselage Mounted Tractor

The fuselage-mounted tractor setup, with a single front propeller, is the best choice for the RC plane. It pulls the plane forward, making the airflow smooth and stable, which helps control, especially for tasks like dropping payloads and steady flight. The front propeller also cools the motor well, important for long flights and high-power movements. This simple design reduces mechanical problems and is easier to maintain. Overall, it provides



balanced performance, making it great for all three missions: payload drops, flying without a payload, and doing inside loops.

4.6.4 Selected Configuration

Table 4-4 Fuselage Mounted Tractor Configuration

				
Parameter	Weight (%)	Tractor	Twin	Pusher
Stability & Control	40	5	4	3
Drag	30	5	3	4
Manufacturability	10	4	3	4
Weight Distribution	20	3	5	4
		4.5	3	2.35

Fuselage mounted tractor is selected as it completes all the missions easily.

4.7 PROPULSION

4.7.1 Single Motor Configuration

A **single-motor configuration** is lighter, simpler, and often more efficient, making it ideal for smaller RC planes or those intended for longer flight times. It requires only one motor, ESC, and propeller, which reduces weight and keeps wiring simple, making it easier to maintain and operate. Single motors provide sufficient thrust for most standard RC planes, particularly for stable cruising and basic aerobatics.

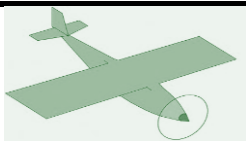
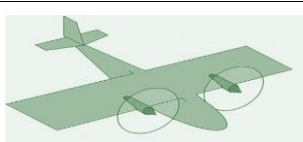


4.7.2 Dual Motor Configuration

A **dual-motor configuration** offers extra thrust and can improve maneuverability and lift, which is beneficial for larger planes, heavy payloads, or complex maneuvers. Dual motors also enable differential thrust, which can aid in tighter turning and yaw control without a rudder. However, this setup is more complex, heavier, and may reduce flight time due to higher power consumption, making it best suited for advanced builds where extra thrust or precise control is necessary.

4.7.3 Selected Configuration

Table 4-5 Motor Configuration

			
Parameter	Weight (%)	Single	Dual
Take Off Length	30	4	5
Propeller Torque	30	4	5
Cruise Efficiency	15	5	3
Weight	25	5	3
		4.4	4.2

The single-motor setup excels in Cruise Efficiency and Weight, making it ideal for efficient, lighter flights. Dual motors perform better for shorter TOFL and stable torque, which is beneficial for heavier planes and complex maneuvers.

4.8 WING MOUNTING CONFIGURATION

4.8.1 Low Wing

Low-wing mounting is sometime preferred to give improved roll response and more agility that are helpful in certain types of aerobatic maneuver. The loss in stability,



however, is an important drawback. When the requirement is a high-precision payload release in an aircraft that could be expected to execute some aerobatics, low wing brings unwanted rolling tendencies, which add complication to control problem at the critical phase like payload release.

4.8.2 Mid-Wing

Mid-wing aircraft offer good aerodynamic efficiency and maneuverability by reducing interference drag, making them suitable for fighter jets and aerobatic planes. However, they have drawbacks, including complex structural integration, reduced internal space for payload or fuel, and more challenging maintenance.

4.8.3 High-Wing

High wing placement provides inherent stability, acting like a pendulum to keep the aircraft level—ideal for controlled flight and precise payload drops. It also offers better ground clearance, reducing the risk of wing damage during takeoff and landing.

4.8.4 Selected Configuration

Table 4-6 Wing Location Configuration

				
Parameter	Weight (%)	High	Mid	Low
Stability & Control	30	5	5	4
Drag	30	3	2	1
Ground Clearance	15	5	3	1
Maneuverability	10	3	3	4
Manufacturability	15	4	1	2
		4.05	3	2.35



High-wing aircraft offer better stability, improved downward visibility, and enhanced ground clearance, making them ideal for rough terrain operations. They also simplify loading due to the lower fuselage, making them popular for cargo, utility, and general aviation.

4.9 FUSELAGE CONFIGURATION

After careful examination, we selected a continuous fuselage design for our aircraft. This configuration ensures superior structural integrity, aerodynamic efficiency, and smooth airflow from nose to tail. It also optimizes space utilization, allowing seamless integration of the cockpit, passenger or cargo compartments, and fuel systems. While modular sections may be incorporated for easier assembly and maintenance, they are joined to function as a unified structure, enhancing the overall performance and reliability of the aircraft.

4.10 WING CONFIGURATION

4.10.1 Elliptical Wing

The Elliptical Wing is one of the most efficient with smooth airflow and a minimum drag profile. This configuration gives good lift-to-drag ratios and improves the flight performance through increased speed and control. This is beneficial for flying without a payload during missions. Its inherent efficiency allows more sustained durations of flight. However, this type of arrangement does demand a precision level of manufacture and tuning that makes this type of complex to manufacture and fine-tune. Such complexity could turn out to be quite an interference when there was the need to prioritize flexibility and versatility in designing an RC plane.



4.10.2 Dihedral Wing

The Dihedral Wing has an advantage about lateral stability because its upward angle helps keep a straight path in flight. This property is best suited for controlled, stable missions such as dropping payloads at a target. A predictable handling from a dihedral wing can reduce the pilot's workload, and this might be beneficial especially for not-so-experienced operators. However, stability is the sacrifice for maneuverability; it is not ideal for maneuvers such as inside loops requiring a higher level of agility and dynamic response.

4.10.3 Delta Wing

The Delta Wing offers high-speed performance and maneuverability, making it ideal for aerobatics like inside loops. Its triangular shape and sturdy structure allow tight turns and rapid movements, essential for precise aerobatic routines. However, this design is less effective for missions requiring slow, accurate flight, such as targeted payload drops. At low speeds, Delta Wing's efficiency drops significantly, making steady flight and precise payload release challenging.

4.10.4 Selected Wing Geometry

Table 4-7 Wing Geometry Configuration

Parameter	Weight (%)	Delta Wing	Elliptical Wing	Tapered Wing
Stability	40	5	4	4
Drag	30	4	3	4
Manufacturability	10	3	3	4
Weight Distribution	20	2	5	4
		3.9	3.8	4



From these types of wings, the multi-mission RC plane design should be tampered with a high placement because a tapered wing is a middle ground between lift and drag. A tapered wing offers flexibility in steady flight as well as aerobatic maneuvers. This wing design should yield reduced induced drag with enhanced aerodynamic efficiency and are important for all three missions: targeted payload drops, flying without payload, and performing inside loops.

4.11 TAIL CONFIGURATION

4.11.1 V-Tail Configuration

V-tails combine horizontal and vertical tails into two angled surfaces, reducing drag and weight, which can improve fuel efficiency and payload capacity. The streamlined design enhances maneuverability, useful for inside loops. However, its control dynamics are complex; mixed yaw and pitch inputs can challenge pilots, especially in tasks requiring high stability, like targeted payload drops. While effective across various wind conditions, V-tails are generally less versatile than other configurations.

4.11.2 T-Tail Configuration

The T-tail configuration places the horizontal stabilizer atop a long vertical fin, providing clean airflow above wing wake and reducing turbulence. This setup is effective for low-speed control, useful in payload drops or steady flight without payload, and offers greater ground clearance, reducing damage risk during rough landings or takeoffs. However, T-tails are less suitable for tight aerobatics or inside loops, with a higher chance of deep stalls if the angle of attack becomes too steep.

4.11.3 Conventional Tail Configuration

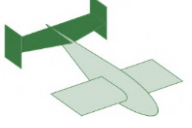
A conventional tail, with a horizontal stabilizer and vertical fin at the rear, offers an ideal design for an RC plane, providing a balance of stability and control. This setup ensures



direct handling, enabling stable flight paths during precise maneuvers like payload drops. The traditional tail configuration provides clear, reliable pitch and yaw control, ensuring consistent maneuverability in almost all flight conditions.

4.11.4 Selected Configuration

Table 4-8 Tail Configuration

				
Parameter	Weight (%)	Conventional	T-Tail	H-Tail
Stability & Control	40	4	5	4
Drag	30	4	3	5
Manufacturability	10	5	4	4
Maneuverability	20	5	3	3
		4.5	3.8	4

Conventional tail is selected as it easily completes all the mission requirements.

4.12 LANDING GEAR

4.12.1 Tricycle Configuration

The **tricycle wheel configuration** features a nosewheel at the front and two main wheels positioned under the fuselage. This design provides better stability, especially during take-off, landing, and taxiing, making it easier to control, particularly for less experienced pilots. The nosewheel configuration reduces the risk of ground loops and provides better handling on paved runways. However, it has lower ground clearance, which increases the risk of propeller strikes on rough or uneven surfaces. The tricycle



configuration is commonly used in modern aircraft due to its ease of handling and overall stability.

4.12.2 Tail dragger Configuration

The **Tail dragger configuration** features a small rear wheel with main wheels positioned forward, offering higher ground clearance and reducing the risk of propeller strikes on rough terrain. It is lighter and simpler in design, often used in off-field operations. However, it requires skilled handling, as the aircraft is more prone to ground loops, particularly in crosswinds, and can be difficult to control during takeoff and landing due to its tail-low attitude. Despite these challenges, the tail dragger excels in rough conditions and is valued for its agility and historical significance.

4.12.3 Selected Configuration

Table 4-9 Landing Gear Configuration

			
Parameter	Weight	Tricycle	Tail Dragger
Take Off Lenth	30	3	4
Payload Access	30	4	3
Weight	20	3	4
Drag	10	3	4
Manufacturability	10	3	4
		3.3	4.6

4.13 MISSION BASED MODEL

We chose a lightweight, low-wing monoplane with a front-mounted propeller for the Design, Build, Fly Competition, designed to handle all three missions well. The front-propeller setup



boosts stability and efficiency for takeoff and landing on the 50 ft runway in Mission 1. Its standard tail helps with smooth landings, earning extra points. The tapered high-wing design increases lift and allows for aerobatic moves like rolls and loops. It also includes a reliable payload system, able to carry 200 grams for Mission 2 and drop it accurately from at least 20 feet for Mission 3. This design combines agility, payload capacity, and stability for top performance in all missions.

4.14 ASSOCIATED UNCERTAINTIES

While our design meets the specified mission requirements, several uncertainties could impact performance. These include variations in environmental conditions, such as wind speed and direction, which may affect flight stability and payload delivery accuracy. Additionally, manufacturing tolerances in the construction of the aircraft may lead to discrepancies in weight and balance, potentially influencing performance during maneuvers and landings. Lastly, the effectiveness of the payload dropping mechanism, particularly when incorporating parachutes, may vary based on design parameters and dynamics during flight. We will conduct thorough testing and simulations to mitigate these uncertainties and enhance our aircraft's performance across all missions.

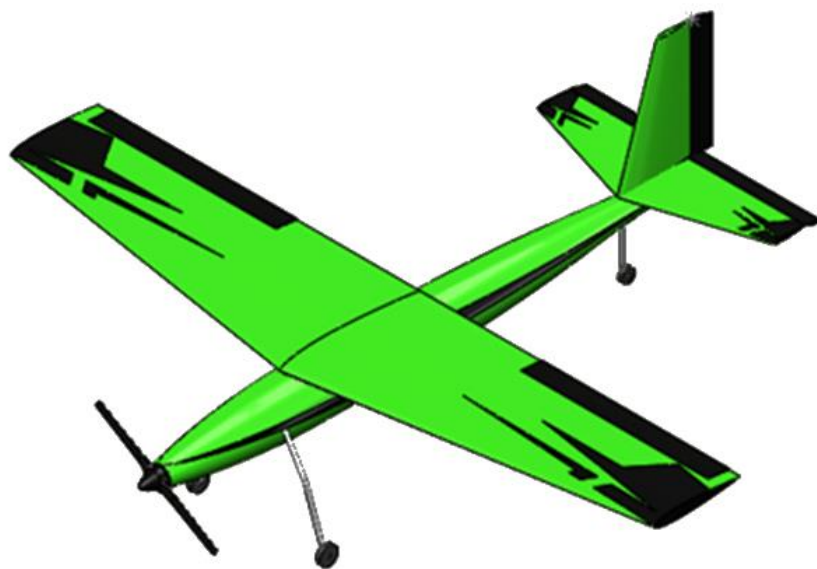


Figure 4-4 CAD Model



5 SELECTION OF AIRFOIL [3]

5.1 AIRFOIL SELECTION

The selection of an airfoil depended on the maximum lift of the airfoil during takeoff and parasitic drag at the design cruise lift coefficient (C_L). High-lift devices were studied to achieve a high $C_{L\ max}$ and minimize TOFL. To select an initial airfoil, a team-developed Python code iterated through airfoils, targeting high $cl\ max$. Candidates were analyzed for their drag at the design lift coefficient, maximum lift to drag ratio, and takeoff performance with flaps.

5.2 MH 112 & E 176 AIRFOIL

The MH 112 and E 176 airfoils are both specialized designs often used in high-performance, low-speed aircraft, such as gliders and UAVs. MH 112 is known for its low drag and high lift characteristics, especially effective at moderate angles of attack, making it well-suited for efficient, stable flight in gliders. The E 176 airfoil, on the other hand, provides a good balance of lift and drag with improved performance in turbulent conditions.

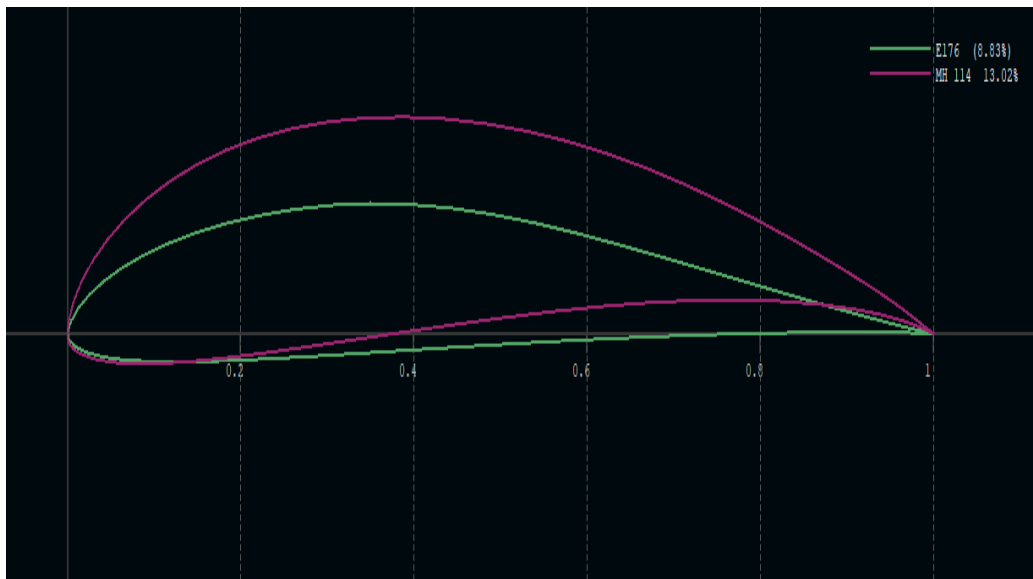


Figure 5-1 MH 112 & E 176 Airfoil



5.3 CL VS CD

This figure shows the lift coefficient C_L versus the drag coefficient C_D . The yellow curves (MH114) generally achieve a higher maximum C_L with slightly lower C_D at higher lift values compared to the magenta curves (E176). This suggests that the MH114 may offer better lift performance with less drag under similar conditions.

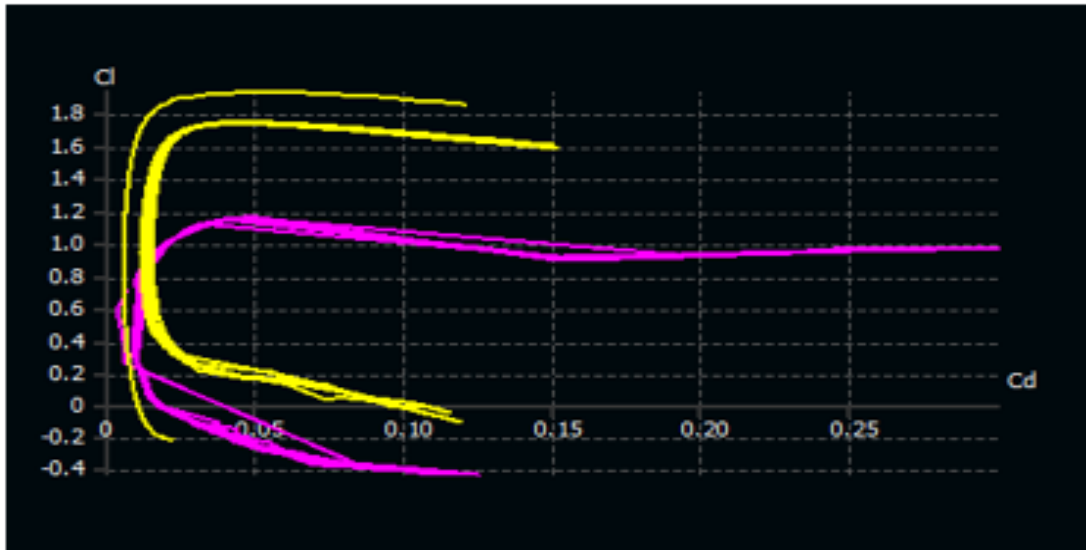


Figure 5-2 CL VS CD

5.4 CL VS ALPHA

This plot illustrates the lift coefficient C_L as a function of the angle of attack (Alpha). The MH114 (yellow) provides a higher lift at low and moderate angles of attack, while the E176 (magenta) shows a more linear increase in C_L before reaching stall. MH114's curves reach a higher peak C_L before stalling, indicating superior lift potential but possibly at the cost of more abrupt stall characteristics.

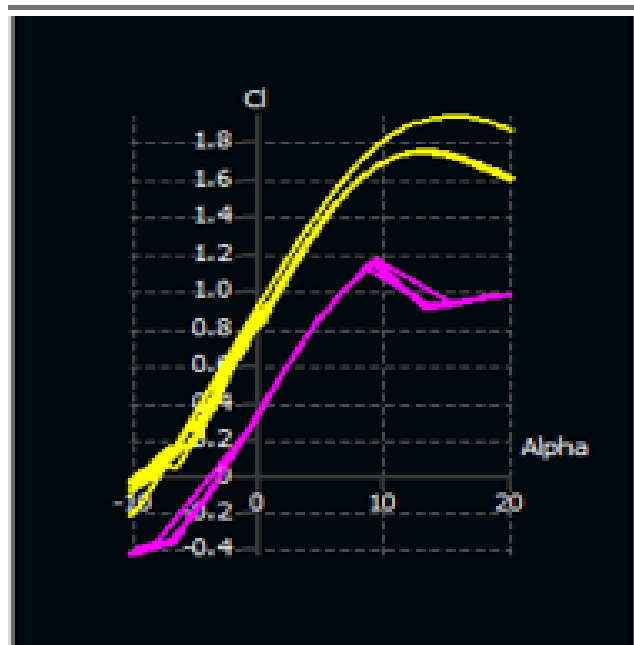


Figure 5-3 CL VS ALPHA

5.5 CM VS ALPHA

This figure shows the moment coefficient C_m against the angle of attack. The E176 (magenta) shows higher C_m values at lower angles, suggesting stronger pitching moments. The MH114 (yellow) generally has lower C_m indicating less pitching moment, which could provide better stability in certain flight conditions.

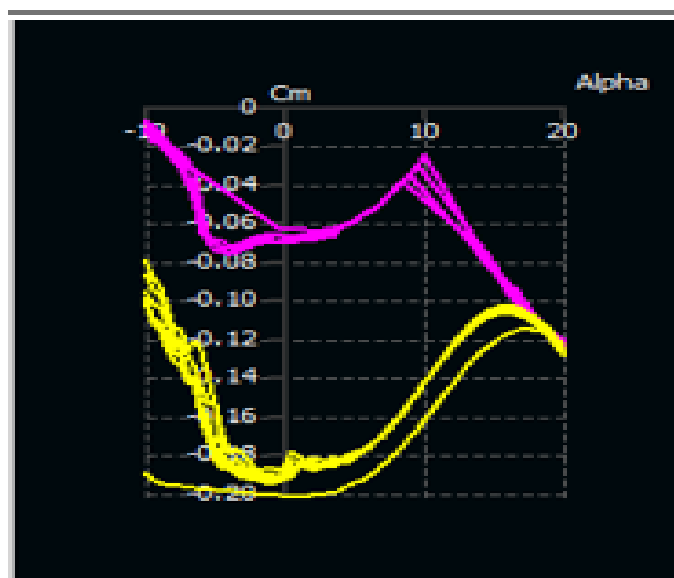


Figure 5-4 Cm vs Alpha Curve



5.6 C_D vs ALPHA

This plot shows the drag coefficient C_D as a function of the angle of attack. MH114 (yellow) has lower C_D values at lower angles, suggesting better efficiency in terms of drag, especially at angles where lift is generated without excessive drag. The lower drag coefficient at these angles suggests that the MH114 can achieve better aerodynamic efficiency, making it ideal for situations where minimizing drag is crucial to maintaining speed and performance.

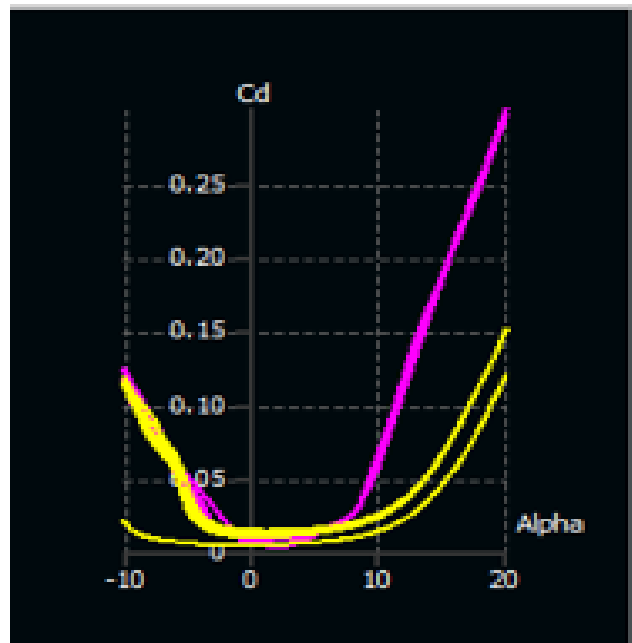


Figure 5-5 C_D vs Alpha

5.7 C_L/C_D vs ALPHA

This figure shows the lift-to-drag ratio C_L/C_D versus the angle of attack, indicating aerodynamic efficiency. The MH114 (yellow) has a higher peak lift-to-drag ratio compared to the E176 (magenta), showing that it is more efficient in producing lift relative to drag, particularly at lower angles of attack.

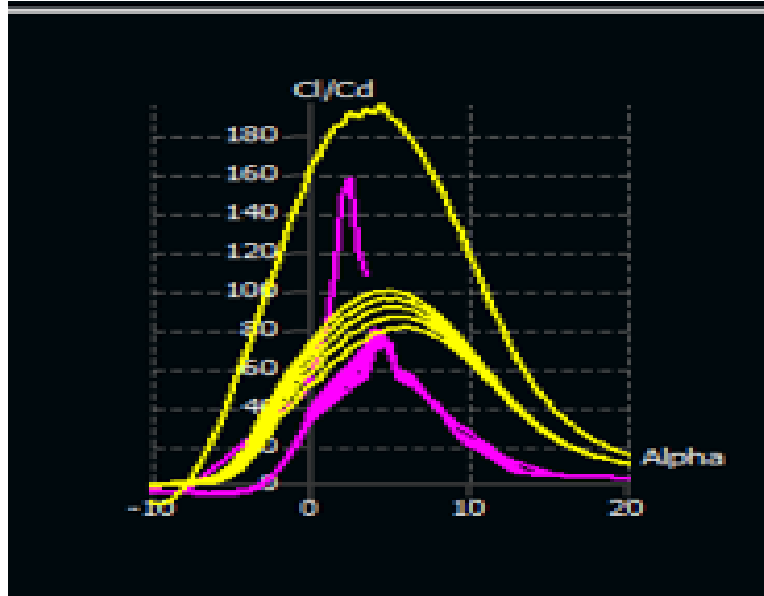


Figure 5-6 Cl/Cd vs Alpha

5.8 SELECTED AIRFOIL

This custom airfoil design combines MH 114 at the root and E 176 at the tip to balance aerodynamic efficiency and strong lift. MH 114 at the root reduces drag and maintains smooth airflow, while E 176 at the tip provides a high lift, especially at low speeds. This setup improves low-speed performance, maneuverability, and stall stability, ideal for gliders, RC planes, and endurance UAVs. Although the cambered tip may add drag at higher speeds, the overall design maintains good lift-to-drag for stable flight and sharp turns.

In summary, this mixed airfoil design offers a tailored solution, optimizing for lift efficiency, control stability, and maneuverability, meeting the specific aerodynamic requirements of advanced flight applications.



6 PRELIMINARY DESIGN [2]

6.1 DESIGN METHODOLOGY

The team collaborated to design all aircraft components to meet the two design objectives: minimizing empty weight and minimizing wingspan. Numerous trade studies were conducted using software simulations and existing models to develop the optimum sizing for all components of the aircraft. Computer models and prototypes allowed for further development of the structural elements of the team's design.

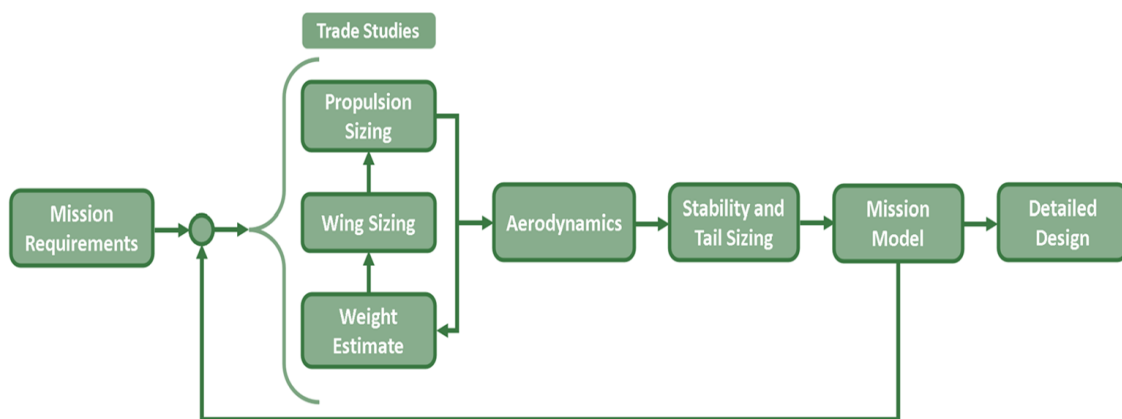


Figure 6-1 Design Methodology

6.2 PRELIMINARY DESIGN METHODOLOGY

In this phase, a preliminary design of the aircraft is made keeping in view the design constraints. Airfoil selection is made based on the design lift coefficient of the aircraft and a detailed analysis of the airfoil is carried out. Geometry sizing is done using Raymer's method and historical trends. Detailed parameters of the designed aircraft along with their values are included in the detailed design section. The illustration below shows the methodology adopted in preliminary design.

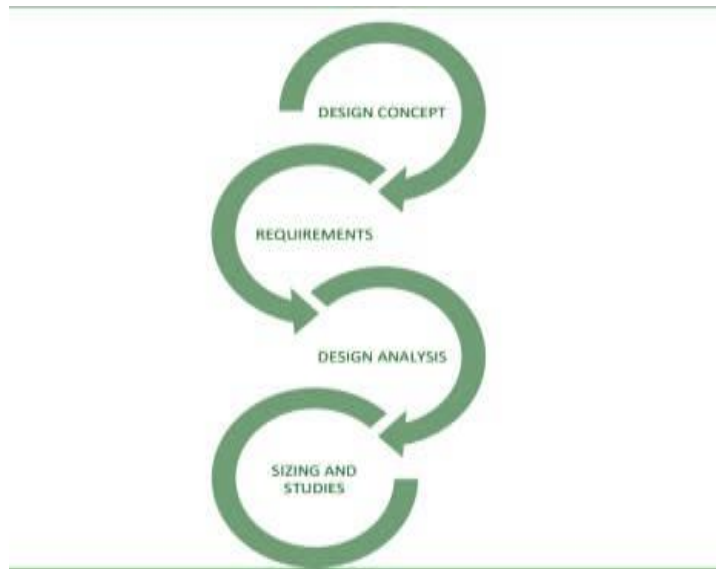


Figure 6-2 Preliminary Design

6.3 PROPULSION SYSTEM

The propulsion subgroup was liable for choosing all the gadgets used to power and control the airplane. Working under the requirements, the group objective was to decide the arrangement of batteries, motor and propellers that would empower the quickest flight times. The propulsion system was likewise expected to meet the extra battery weight and current limit necessities. The group considered picking a framework that would give high static thrust while empowering a high cruise speed.

Table 6-1 Propulsion System

Parameter	Motor	ESC	Battery	Receiver	Servo
Specifications	EMAX GT 2826/04	Skywalker 60A	GENS ACE	R2006 GS	Perfex KS-12 MG

6.4 GEOMETRY SIZING

“Sizing” is the most important calculation in aircraft design, more so than drag, or stress, or even cost. Sizing determines the size of the aircraft, specifically the weight that an aircraft must possess to perform a certain mission while carrying its intended Payload i.e. Gross Take-off Weight.



6.5 WING SIZING

Wing sizing plays a critical role in aircraft design as it directly impacts lift, stability, and fuel efficiency. Accurate calculations ensure that the wing can generate enough lift to support the aircraft's weight, particularly during takeoff and landing. Key parameters like wingspan, aspect ratio, and wing loading are optimized to balance performance and aerodynamic efficiency. Effective wing sizing also improves cruise performance, maneuverability, and fuel consumption, ensuring the aircraft meets mission requirements while operating efficiently.

Table 6-2 Wing Parameter

Parameter	Unit (meter)
Wing Area	0.314 sq meter
Wingspan	1.395 meters
Taper ratio	0.75
Root chord	0.257 meters
Tip Chord	0.192 meters
MAC	0.226 meters

6.5.1 Wing Area

Wing area can be calculated by using the wing loading formula:

$$\text{Wing loading} = \frac{W}{S}$$

The weight of the aircraft is 2.2 kg, we can estimate the wing area by using a typical wing loading value of 6 kg/m². Since the typical value of wing loading for RC planes is between 5-10 kg/m².

So,

$$\frac{W}{S} = \frac{2.2}{S}$$



$$7 = \frac{2.2}{S}$$

$$S = 0.314 \text{ m}^2$$

6.5.2 Aspect Ratio

The aspect ratio of an aircraft is a measure of how long its wing is to how large its wing surface area is. It is generally given by the expression:

$$AR = \frac{b^2}{S}$$

b = Wingspan

S = Wing Surface Area

The aspect ratio lies between 5-8 for RC planes. 8:1 is common for gliders with built-up balsa wings, and high-end carbon fiber gliders are usually more like 12:1. The limit for foam or posterboard construction is probably 7:1 with a well-sorted design, but something in the 5:1 or 6:1 range is easier to build and more likely to be successful. So,

Aspect Ratio = 6.2

Now,

$$b = \sqrt{AR * S}$$

$$b = \sqrt{6.2 * 0.314}$$

$$b = 1.395 \text{ m}$$

6.5.3 Taper Ratio

Taper Ratio is the ratio of tip chord to the centerline root chord. The taper ratio affects the distribution of lift across the span of the wing. As the taper ratio increases, the wing planform starts resembling an elliptical planform meaning that induced drag decreases.

The taper ratio of the reference aircraft is taken for initial sizing.



$$\lambda = \frac{C_{\text{tip}}}{C_{\text{root}}}$$

For Taper Wing,

$$\lambda = 0.75$$

6.5.4 Root Chord

$$C_{\text{root}} = \frac{2 * S}{b * (1 + \lambda)}$$

$$C_{\text{root}} = \frac{2 * 0.314}{1.395 * (1 + 0.75)}$$

$$C_{\text{root}} = 0.257 \text{ m}$$

6.5.5 Tip Chord

$$C_{\text{tip}} = C_{\text{root}} * \lambda$$

$$C_t = (0.257) * (0.75)$$

$$C_t = 0.192 \text{ m}$$

6.5.6 Mean Aerodynamic Chord

$$c = \frac{2}{3} * C_{\text{root}} * \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

$$c = \frac{2}{3} * (0.259) * \frac{1 + 0.75 + 0.75^2}{1 + 0.75}$$

$$c = 0.226 \text{ m}$$

6.6 FUSELAGE SIZING

Fuselage is an aircraft's main body section that holds crew and passengers or cargo.

Its sizing can be done using some initial estimation.



Table 6-3 Fuselage Parameter

Parameter	Unit (meter)
Fuselage Length	1.0462meters
Fuselage Height	0.104meters
Fuselage Depth	0.104meters

6.6.1 Fuselage Length

By using Rule of Thumb, Fuselage length is 70% to 75% of wingspan. We take 75% of the wingspan for RC aircraft.

$$\text{Fuselage length} = \frac{75}{100} * 1.395$$

$$\text{Fuselage length} = 1.046 \text{ m}$$

6.6.2 Fuselage Width & Height

Fuselage width and height is 10% of fuselage length.

$$\text{Fuselage width} = \frac{10}{100} * 1.046 \text{ m}$$

$$\text{Fuselage width} = 0.1046 \text{ m}$$

Similarly

$$\text{Fuselage height} = 0.1046 \text{ m}$$

6.7 EMPENNAGE SIZING

Empennage controls the movements of aircraft in pitch and yaw-axis. This part of the aircraft consists of different configurations based upon the required design. The Horizontal and Vertical stabilizers make up the Empennage.



Table 6-4 Raymer Table for HT & VT Sizing

	Horizontal Tail		Vertical Tail	
	Δ	λ	Δ	λ
Fighter	3-4	0.2-0.4	0.6-1.4	0.2-0.4
Sailplane	6-10	0.3-0.5	1.5-2.0	0.4-0.6
Others	3-5	0.3-0.6	1.3-2.0	0.3-0.6
T-tail	-	-	0.7-1.2	0.6-1.0

6.7.1 Horizontal Stabilizers

The length of Horizontal Stabilizer is calculated using the following formula:

$$V_H = \frac{S_H * L_H}{S * MAC}$$

V_H = Horizontal Tail Volume Coefficient

V_L = Vertical Tail Volume Coefficient

L_H = Moment arm distance between CG and leading edge of the Horizontal Stabilizer.

S_H = Unknown Area of the Horizontal Stabilizer.

S_V = Unknown Area of the Vertical Stabilizer.

MAC = Mean Aerodynamic Chord.

S = Wing Area.

L_H = 48% of the Fuselage Length

$$L_H = \frac{48}{100} * 1.0125$$

$$L_H = 0.486 \text{ m}$$

We know that

$$L_H = L_V$$



$$V_H = 0.40$$

$$V_v = 0.035$$

Table 6-5 Horizontal Tail Parameter

HT Parameter	Unit (meter)
HT Area	0.058 sq meters
HT-span	0.512 meters
Taper ratio	0.45
HT Root chord	0.157 meters
HT Tip Chord	0.070 meters
HT MAC	0.119 meters

6.7.1.1 Horizontal Tail Area

$$S_H = \frac{V_H * S * MAC}{L_v}$$

$$S_H = \frac{(0.40) * (0.314) * (0.226)}{0.482}$$

$$S_H = 0.0584 \text{ m}^2$$

HORIZONTAL TAIL PARAMETER

The dimensions of the Horizontal tail parameter can be found using aspect ratio of 4 and 0.45 and Taper ratio of 0.3 and 0.6 respectively. So,

$$AR = 4.5$$

Similarly,

$$\lambda = 0.45$$

6.7.1.2 Horizontal Stabilizer Span

$$AR = \frac{b^2}{S}$$



$$b = \sqrt{AR * S_H}$$

$$b = \sqrt{4.5 * (0.0584)}$$

$$\mathbf{b = 0.512 \text{ m}}$$

6.7.1.3 Root Chord

$$C_r = \frac{2 * S}{b * (1 + \lambda)}$$

$$C_r = \frac{2 * 0.0584}{0.512 * (1 + 0.45)}$$

$$\mathbf{C_r = 0.157 \text{ m}}$$

6.7.1.4 Tip Chord

$$C_t = C_r * \lambda$$

$$C_t = (0.157) * (0.45)$$

$$\mathbf{C_t = 0.0706 \text{ m}}$$

6.7.1.5 Mean Aerodynamic Chord

$$c = \frac{2}{3} * c_r * \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

$$c = \frac{2}{3} * (0.157) * \frac{1 + 0.45 + 0.45^2}{1 + 0.45}$$

$$\mathbf{c = 0.119 \text{ m}}$$



6.7.2 Vertical Stabilizer Parameter

Table 6-6 Vertical Tail Parameter

VT Parameter	Unit (meter)
VT Area	0.031 sq meter
VT-span	0.25 meters
Taper ratio	0.45
VT Root chord	0.173 meters
VT Tip Chord	0.078 meters
VT MAC	0.131 meters

6.7.2.1 Vertical Tail Area

$$S_V = \frac{V_H * S * MAC}{L_V}$$

$$S_V = \frac{(0.40) * (0.314) * (0.226)}{0.482}$$

$$S_V = 0.0584 \text{ m}^2$$

6.7.2.2 Vertical Tail Parameter

The dimensions of the Vertical tail parameter can be found using aspect ratio of 1.3 and 2 respectively. So,

$$AR = 2$$

Similarly

$$\lambda = 0.45$$

6.7.2.3 Vertical Tail Span

$$AR = \frac{b^2}{S}$$



$$b = \sqrt{AR * S_v}$$

$$b = \sqrt{4 * (0.0315)}$$

$$\mathbf{b = 0.250\ m}$$

6.7.2.4 Vertical Tail Chord

$$AR = \frac{\text{Span}}{\text{Chord}}$$

$$\text{Chord} = \frac{\text{Span}}{AR}$$

$$\text{Chord} = \frac{0.250}{2}$$

$$\mathbf{\text{Chord} = 0.032\ m}$$

6.7.2.5 Root Chord

$$C_r = \frac{2 * S}{b * (1 + \lambda)}$$

$$C_r = \frac{2 * 0.0315}{0.250 * (1 + 0.45)}$$

$$\mathbf{C_r = 0.173\ m}$$

6.7.2.6 Tip Chord

$$C_t = C_r * \lambda$$

$$C_t = (0.173) * (0.45)$$

$$\mathbf{C_t = 0.0778\ m}$$

6.7.2.7 Mean Aerodynamic Chord

$$c = \frac{2}{3} * c_r * \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

$$c = \frac{2}{3} * (0.184) * \frac{1 + 0.45 + 0.45^2}{1 + 0.45}$$



$$c = 0.131 \text{ m}$$

6.7.3 Control Surface Sizing

Table 6-7 Control Surface Sizing

Control Surface	Unit (meter)
Aileron	
Aileron-span	0.837 meters
Aileron chord	0.045 meters
Elevator	
Elevator span	0.512 meters
Elevator chord	0.03 meters
Rudder	
Rudder span	0.250 meters
Rudder chord	0.032 meters

6.7.3.1 Rudder Span

To calculate the span of rudder, we take 100% of vertical tail span, So

$$b = 0.250 \text{ m}$$

6.7.3.2 Rudder Chord

The dimensions of the Rudder chord can be found using 15% to 25% of Vertical Tail Mean Aerodynamic Chord respectively. So,

$$c = \frac{20}{100} * 0.131$$

$$c = 0.032 \text{ m}$$

6.7.3.3 Elevator Span

To calculate the span of elevator, we take 100% of horizontal tail span, So

$$b = 0.512 \text{ m}$$



6.7.3.4 Elevator chord

The dimensions of the elevator chord can be found using 15% to 25% of Vertical tail Mean Aerodynamic Chord respectively. So,

$$c = \frac{20}{100} * 0.1282$$
$$\mathbf{c = 0.03\ m}$$

6.7.3.5 Aileron Span

To calculate the span of Aileron, we take 50 to 90% of wingspan, So we take 60%.

$$b = \frac{60}{100} * 1.395$$
$$\mathbf{b = 0.837\ m}$$

6.7.3.6 Aileron Chord

The dimensions of the aileron chord can be found using 15% to 25% of Wing Mean Aerodynamic Chord respectively. So,

$$c = \frac{20}{100} * 0.226$$
$$\mathbf{c = 0.0452\ m}$$



7 DETAILED DESIGNING

7.1 DIMENSIONAL PARAMETERS OF FINAL DESIGN

The table shows the main aircraft dimensions and characteristics the team deemed ideal for the 2024 AIAA Design, Build, Fly competition.

Table 7-1 Detailed Design

Wing		Tail	
Airfoil	MH 114 & E 176	Airfoil	NACA 0012
Span, b	4.59 ft (1.4 m)	Horizontal Span	20.16 in (0.512 m)
MAC	0.74 ft (0.226 m)	Horizontal Chord	4.68 in (0.119 m)
Planform Area, S	3.38 ft ² (0.314 m ²)	Vertical Span	13.0 in (0.250 m)
Aspect Ratio, AR	6.2	Vertical Chord	5.16 in (0.131 m)
Incidence Angle	-2°	Horizontal Tail Area	1.63 ft ² (0.151 m ²)
Static Margin	16-20%	Incidence Angle	0°
Fuselage		Vertical Tail Area	90.5 in ² (0.0584 m ²)
Total Length	41.18 in (1.046 m)	Controls and Power	
Width	4.118 in (0.1046 m)	Receiver	R2006 GS
Maximum Height	4.118 in (0.1046 m)	Servos	Perfex KS-12 MG
Rudder Span	9.84 in (0.250 m)	Battery Model	Spektrum Smart G2
Rudder Chord	1.2 in (0.032 m)	Cell Count	3S
Motor		Peak Voltage	29.6 V
Model	Scorpion A-4220	Pack Weight	1.59 lbm (0.72 kg)
kV	540	Propeller	
Power Rating	2553 Watts	Manufacturer	APC Propellers
No-Load Current, I_0	1.54 Amps	Mission 1 (M1)	11x7S & 11x7P in
Internal Resistance	18.2 mΩ	Mission 2 (M2)	11x7S & 11x7P in
Weight	10.16 oz (288 g)	Mission 3 (M3)	11x6S & 11x6P in



7.2 STRUCTURAL CHARACTERISTICS AND PERFORMANCE

7.2.1 Wing

The wing design uses Thermo-col sheets for the outer skin, offering lightweight strength to handle aerodynamic forces, while balsa wood spars support the main structural loads. Reinforcements along the leading and trailing edges increase rigidity, preventing deformation. This semi-monocoque structure combines Thermo-col and balsa, efficiently distributing forces. The aerodynamic shape and spar placement improve airflow, enhancing lift-to-drag ratio, stability, and performance at various speeds. This design is lightweight, cost-effective, and meets both structural and budget requirements

7.2.2 Fuselage

The outer cover is made from a light outer layer of thermos-col. This has been selected because of low density and the property that it will be a good insulator of heat along with contribution to structural strength. The shape of the fuselage can maintain its under aerodynamic loads, so the aircraft can remain very light in weight. Longerons and stringers form the skeleton structure of an aircraft with balsa wood. The longerons are arranged longitudinally within the fuselage to oppose bending forces and distribute loads uniformly.

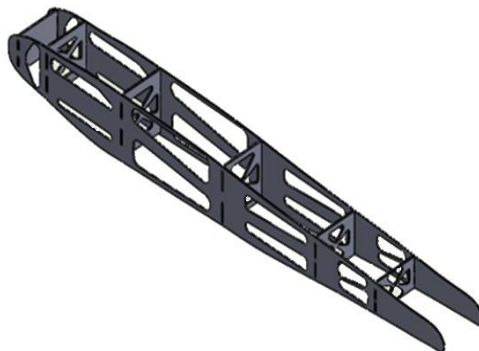


Figure 7-1 Internal Structure of Fuselage



A single Balsa wood and carbon fiber rod has been added to the fuselage to enhance further the structure. The Balsa wood rod runs longitudinally along the centerline of the fuselage, enhancing rigidity and distributing stress better all over the frame. The rod ensures that the fuselage can withstand bending forces and dynamic loads, especially at places where the structure tends to flex, such as takeoff, landing.

The thermos-col outer layer, along with a balsa wood framework and carbon fiber rod, has been used. The following is a semi-monocoque structure and creating optimum weight distribution as well as strength. Adding these critical support elements without carrying undue weights to the fuselage ensured it remains stable and is less prone to deformation while maintaining light weight. A judicious choice of the above combination helped my team in the preparation of this very robust, efficient, and cost-effective fuselage to meet all demands posed by the competition

7.2.3 Empennage

The tail section is made from thermos-col, chosen for its lightweight and durable properties essential to this application. Its streamlined design reduces drag, enhancing flight performance, while its structure resists aerodynamic forces during maneuvers for added stability. Thermo-col's stiffness and flexibility improve control over pitch and yaw, allowing for responsive handling. The tail's shape also optimizes airflow, boosting lift without complicating manufacturing. Its single-material design allows for easy integration with the fuselage and wings

7.3 PAYLOAD DROPPING MECHANISM

Three payloads can be released from an RC plane using this payload dropping method, which employs two servos. Payloads 1 and 2 are released by Servo 1, and Payload 3 is released by Servo 2. With both manual and automatic control options, the system is built to provide exact control over the timing and manner of payload drops. All cargoes are first firmly fastened. When Servo 1 is triggered, it moves to release Payloads



1 and 2 either simultaneously or sequentially. Payload 3 is controlled by Servo 2, which releases it upon instruction. Flexible release timing is made possible by the switches that the pilot or the RC plane's flight controller uses to activate the servos.

Accidental drops are avoided by safety features like fail-safe switches and locking pins. A pre-flight inspection guarantees that the system is operating correctly and that all payloads are firmly attached. Purposes for this technique include military operations, environmental monitoring, and the delivery of humanitarian relief. In mission-critical situations, the system guarantees dependable payload release by offering precise control and safety measures.

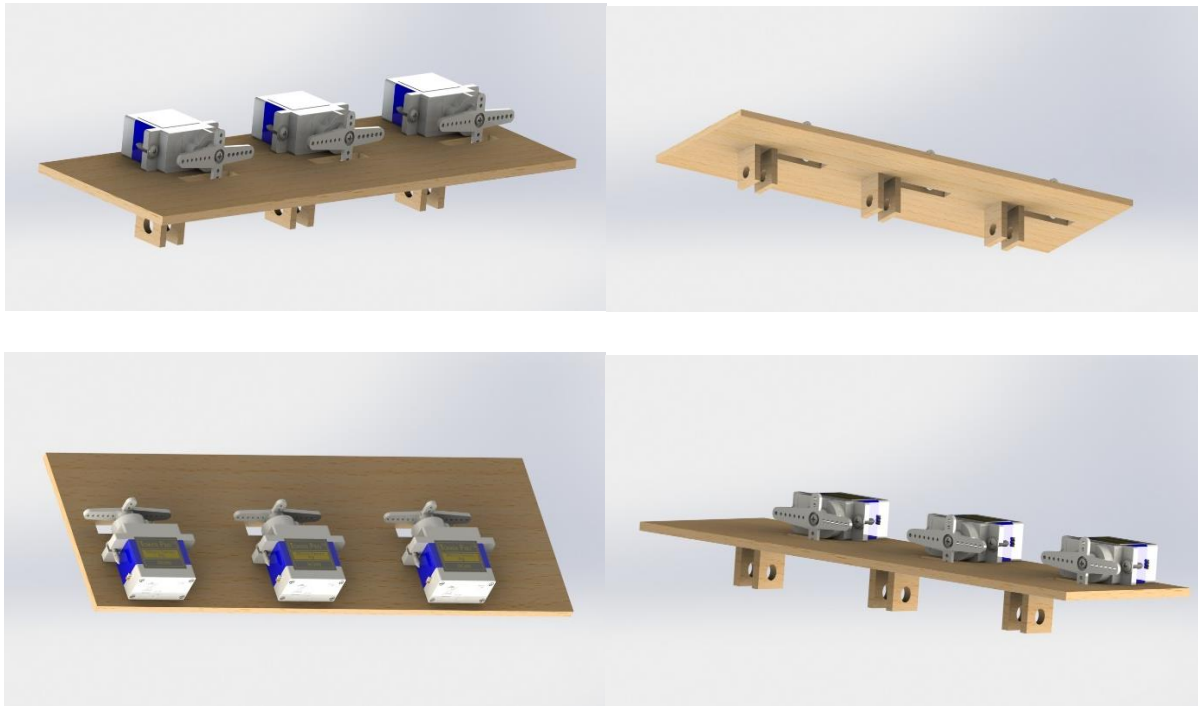


Figure 7-2 Payload Dropping Mechanism



7.4 TRANSMITTER

The 6J Transmitter is an RC system that consists of six channels and uses the S-FHSS/FHSS 2.4 GHz frequency band, providing secure and interference-free communication to the RC plane. The two-stick controller features perfect control over all important flight operations like throttle, ailerons, elevator, rudder, and additional channels of payload release or specific flight maneuvers.



Figure 7-3 Transmitter

7.5 RECEIVER

The R2006GS Receiver is a lightweight, compact unit that ensures reliable communication for RC planes, supporting S-FHSS and FHSS protocols with auto-detection for stable signal reception. Its small size (25 x 43.1 x 8.8 mm) and light weight (8.5 g) make it easy to fit into the plane without adding bulk. It operates from 4.8V to 7.4V, compatible with various power setups, ensuring smooth and precise flight control.



Figure 7-4 R2006GS Receiver



7.6 SERVO MOTOR

The Perfex KS-12MG servo from Kyosho is a high-performance, reliable part to use when controlling the main flying surfaces of the RC plane. Its metal gears make the servo very reliable and constant in performance, giving it the edge it needs to carry out precision moves during DBFC missions that include payload drops and aerobatic loops. High torque capacity at 8.5 kg-cm is adequate in handling the loads caused by aerodynamics to give stability.



Figure 7-5 Perfex KS-12MG

7.7 BATTERY

The GENS ACE LiPo battery is designed with the intention to provide excellent efficient and reliable power sources to the RC planes. Further features of smart technology guarantee secure charging and discharging operations and provide the plane with all the energy in required high-performance flight situations. Its high discharge of 50C ensures proper consistency of power supply for demanding maneuvers and payload activities.



Figure 7-6 Battery 5500 MAH

7.8 MOTOR

The EMAX GT2826 1100KV brushless motor is powerful and efficient, generating 3100 grams of thrust. Its 1100KV rating makes it ideal for aerobatic flying and payload delivery. The motor's strong, lightweight design enhances performance, agility, and the power-to-weight ratio in RC planes.



Figure 7-7 EMAX GT2826 1100KV



7.9 ELECTRONIC SPEED CONTROLLER (ESC)

The Hobby wing Skywalker 60A Brushless ESC with UBEC 5V 5A is a top-quality electronic speed controller for RC aircraft, designed to enhance performance. It supports a variety of brushless motors, delivering reliable power and precise control, ideal for competitive flying. Its integrated UBEC ensures stable operation and efficient power distribution in demanding flight conditions, such as in DBFC competitions.



Figure 7-8 Hobby wing Skywalker 60A

7.10 CONTROLLER

The Speedy Bee F405 Wing flight controller is an essential component of this RC aircraft, chosen for its ability to provide advanced flight modes, stability, and autonomy. While traditional RC planes can be flown manually, incorporating this flight controller enables precise control and automation, critical for missions involving payload delivery and efficient navigation.



Figure 7-9 Controller

This flight controller was selected due to its cost-effectiveness, modern features, and ability to support advanced functionalities needed for this project. It allows seamless integration of manual and autonomous control, ensuring operational flexibility and reliability.

7.10.1 Working of the Flight Controller

The flight controller acts as the aircraft's brain, integrating data from multiple onboard sensors, including:

- 1) **Accelerometer and Gyroscope:** For determining orientation and movement.
- 2) **Compass:** For heading information.
- 3) **GPS:** For position and navigation.
- 4) **Airspeed Sensor:** For real-time speed data.

The collected sensor data is processed through an **Extended Kalman Filter (EKF)** to provide accurate state estimation. It also takes input from the pilot via the transmitter and combines it with sensor data to generate precise control signals sent to the servos, ensuring stable flight and responsiveness to commands.

The flight controller runs on the **ArduPilot software platform**, which has been customized for the specific requirements of this project, allowing optimization for mission-specific tasks such as payload delivery and waypoint navigation.



7.10.2 Flight Modes

Key flight modes utilized in this project include:

- 1) **FBWA (Fly By Wire A):** A stabilized mode where the aircraft assists the pilot by maintaining level flight while allowing controlled maneuvers.
- 2) **AUTO:** Fully autonomous mode for waypoint-based missions, used for executing precise payload drops.
- 3) **RTL (Return to Launch):** A safety mode where the aircraft automatically returns to the launch point in case of signal loss or on command.

These modes enable efficient and safe operation, allowing the plane to perform tasks that are challenging with manual control alone.

7.10.3 Flight Data Recording and Post-Flight Analysis

The flight controller also records all flight data, including sensor readings, control inputs, and flight path. This data can be visualized and analyzed post-flight, providing valuable insights for mission assessment and performance optimization. The ability to replay flights aids in diagnosing issues, refining mission planning, and improving future operations.



7.11 RC PLANE ELECTRONICS COMPONENTS SPECIFICATION

7.11.1 Transmitter Specifications

Table 7-2 Transmitter Specification

Transmitter Specifications		
1	Brand	Futaba
2	Model	R2006GS (T6J compatible Receiver)
3	Channel	6 Channel
4	Voltage Range	3.5V --- 8.4V
5	Dimension	40mm ×25mm ×15mm
6	Weight	Around 9g

7.11.2 Receiver Specifications

Table 7-3 Receiver Specifications

Receiver Specifications		
1	Brand	Futaba
2	Model	T6J
3	Channel	6 Channel
4	Voltage Range	6.0V --- 8.4V
5	Dimension	176mm ×182mm ×88mm
6	Weight	Around 550g



7.11.3 Servo Motor Specifications

Table 7-4 Servo Motor Specifications

Servo Motor Specifications		
1	Brand	Kysoho Perfex KS 12-MG
2	Type	Analog Servo
3	Motor Type	Coreless
4	Voltage Range	4.8V --- 6V
5	Dimension	40mm ×20mm ×37mm
6	Weight	56g

7.11.4 Battery Specifications

Table 7-5 Battery Specifications

Battery Specifications		
1	Brand	GENS ACE
2	Capacity	5500 mAh
3	Nominal Voltage	11.1V
4	Discharge Rate	45C
5	Dimension	38mm ×42mm ×20mm
6	Weight	270g --- 290g

7.11.5 Motor Specifications

Table 7-6 Motor Specifications

Motor Specifications		
1	Model	EMAX GT 2826 1100KV
2	KV Rating	1100KV
3	Stator Size	28mm ×26mm
4	Motor Diameter	35.2mm
5	Motor Length	40.7mm
6	Propeller Size	10 ×5 --- 12×6



7.11.6 Electronic Speed Controller Specifications

Table 7-7 Electronic Speed Controller (ESC) Specification

Electronic Speed Controller (ESC) Specifications		
1	Brand	Kysoho Perfex KS 12-MG
2	Type	Analog Servo
3	Motor Type	Coreless
4	Voltage Range	4.8V --- 6V
5	Dimension	40mm ×20mm ×37mm
6	Connector Type	Standard Three Pin



8 COMPUTER AIDED DESIGNING (CAD) MODEL [4]

8.1 ABOUT CAD MODEL

An aircraft's CAD (Computer-Aided Design) model is essential for a few reasons. It helps to see and analyze the structure of the aircraft by providing a precise and comprehensive depiction of its design. The assessment of aerodynamic performance, weight distribution, and structural integrity is made easier by CAD models, which promotes the safe and effective construction of aircraft. They help engineers save time and money by allowing them to tweak and optimize designs prior to actual prototyping. CAD models are also necessary for diverse teams' documentation, cooperation, and communication throughout the aircraft's whole lifespan, from design to maintenance.

8.2 AIRCRAFT CAD MODEL

8.2.1 3-D View



Figure 8-1 3D Rendered View



8.2.2 Front View

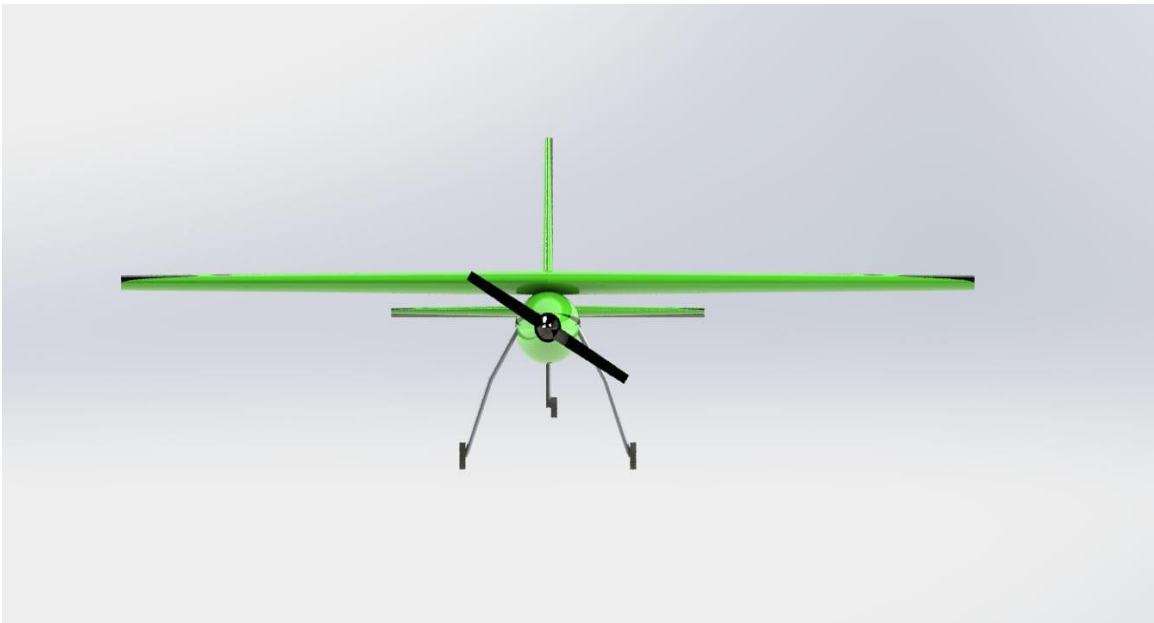


Figure 8-2 Front View

8.2.3 Side View

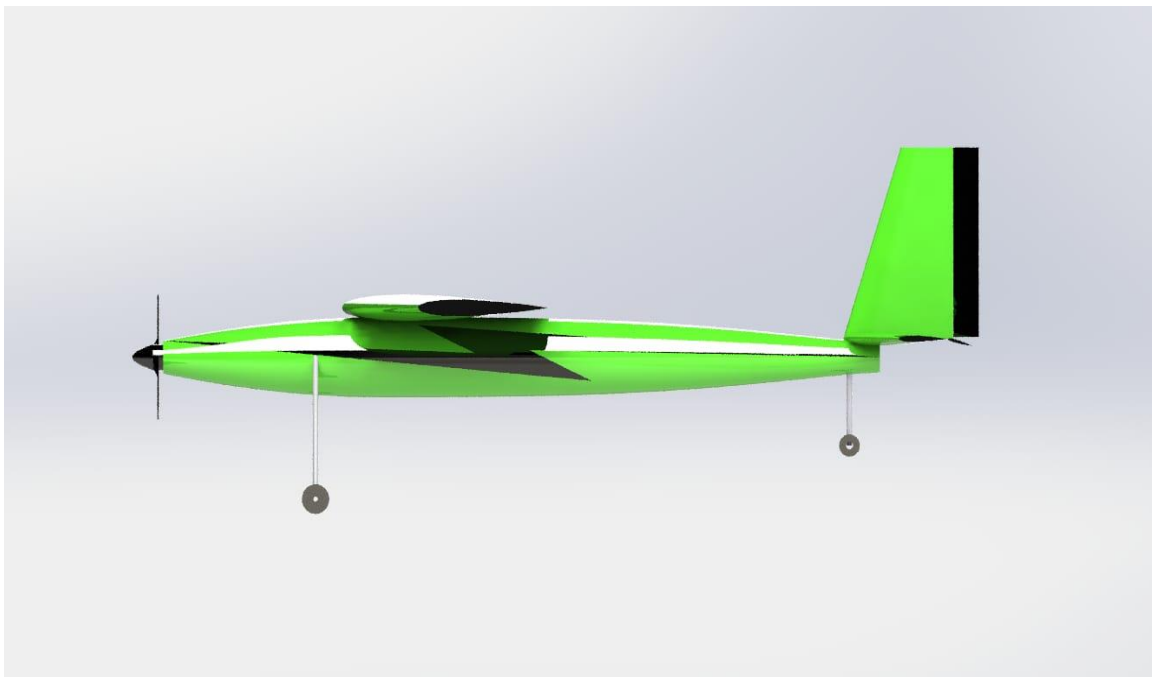


Figure 8-3 Side View



8.2.4 Top View

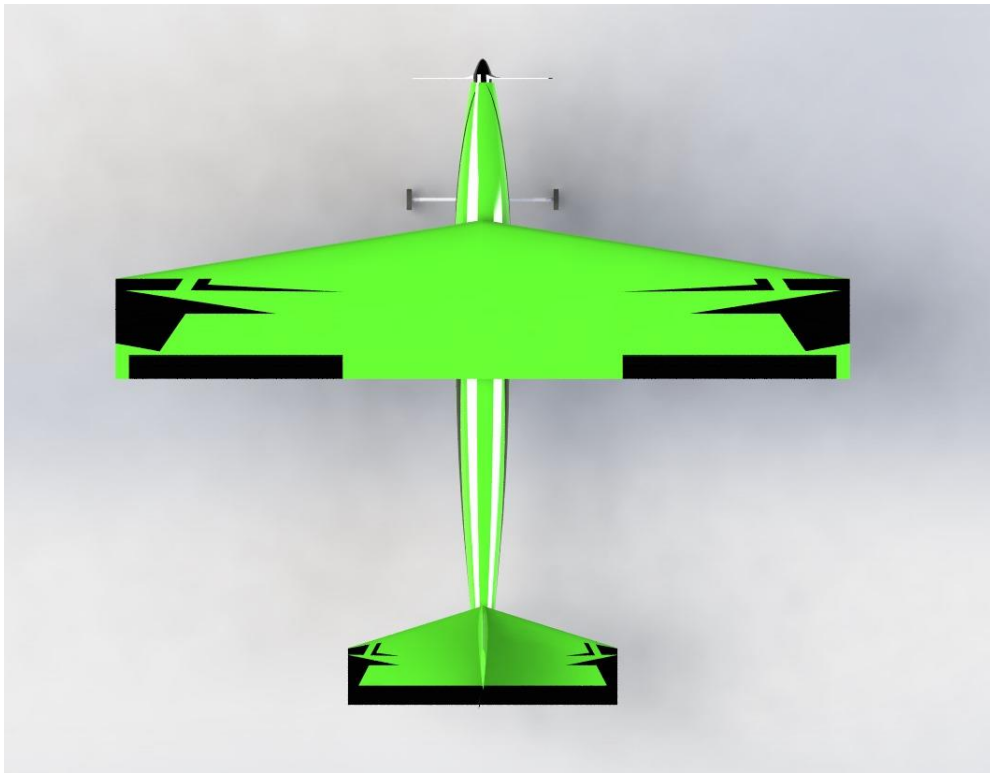


Figure 8-4 Top View

9 COMPUTER AIDED DESIGN (CAD) DRAWING [4]

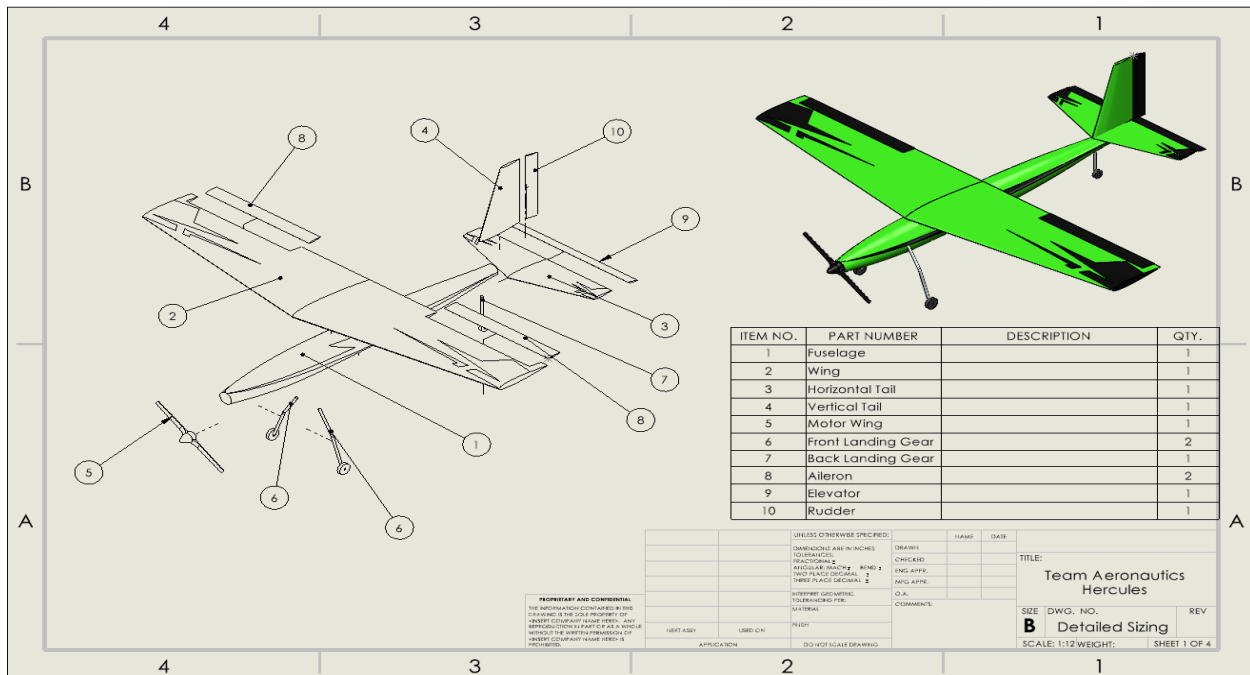


Figure 9-1 Detailed Sizing on Sheet 1

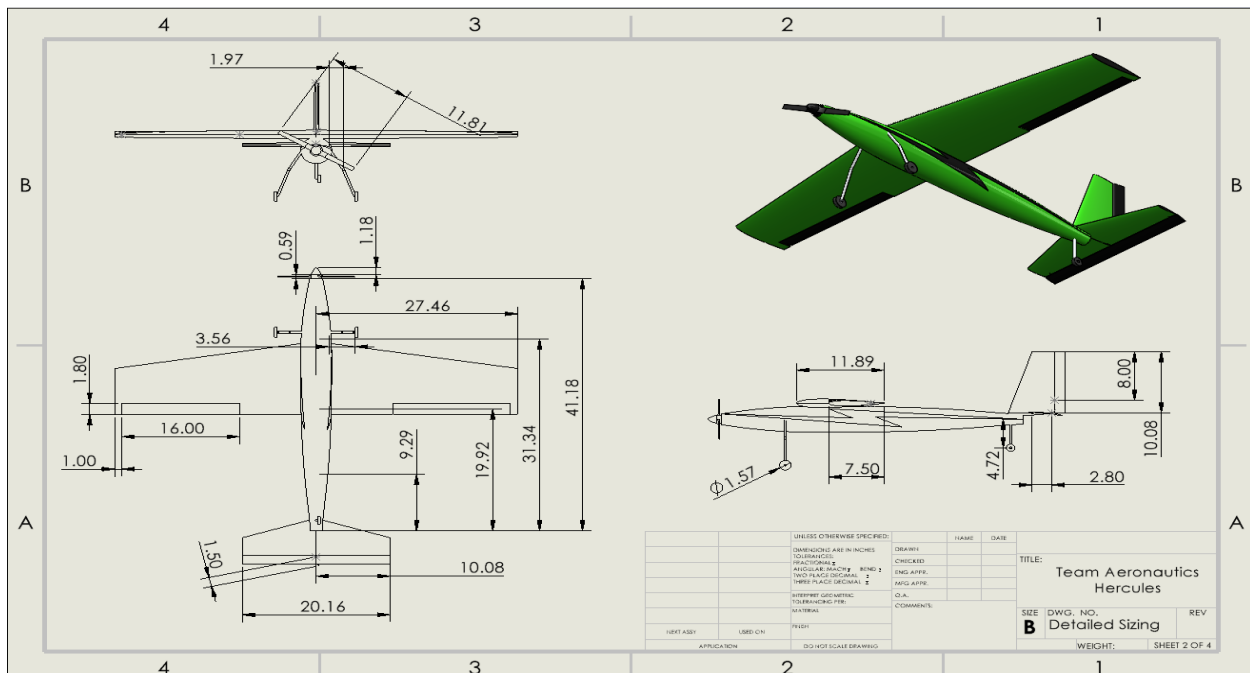


Figure 9-2 Detailed Sizing on Sheet 2

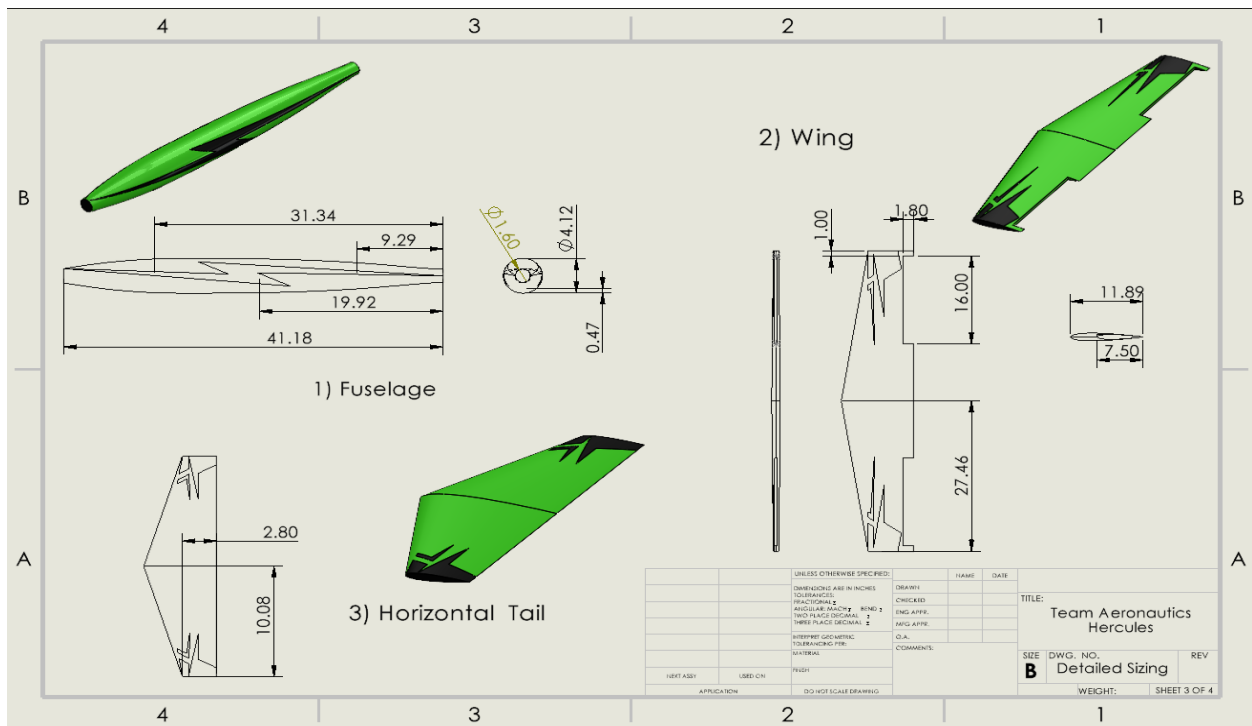


Figure 9-3 Detailed Sheet Sizing of sheet 3

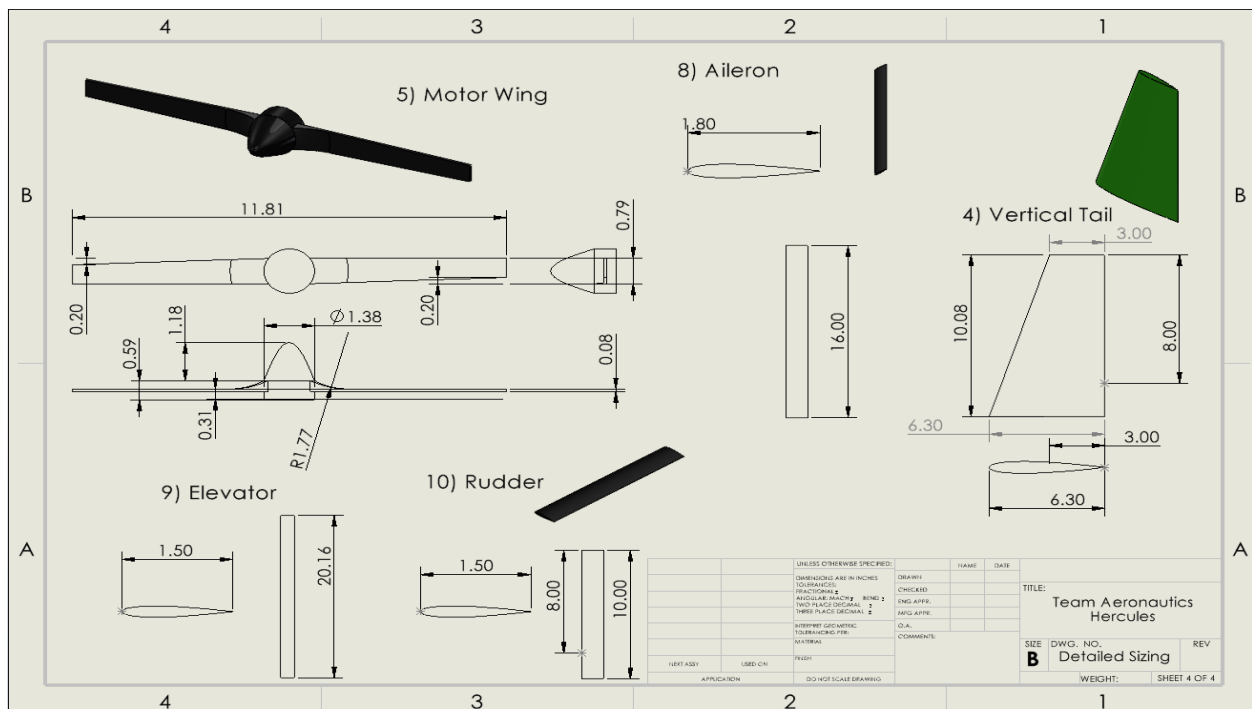


Figure 9-4 Detailed Sizing of Sheet 4

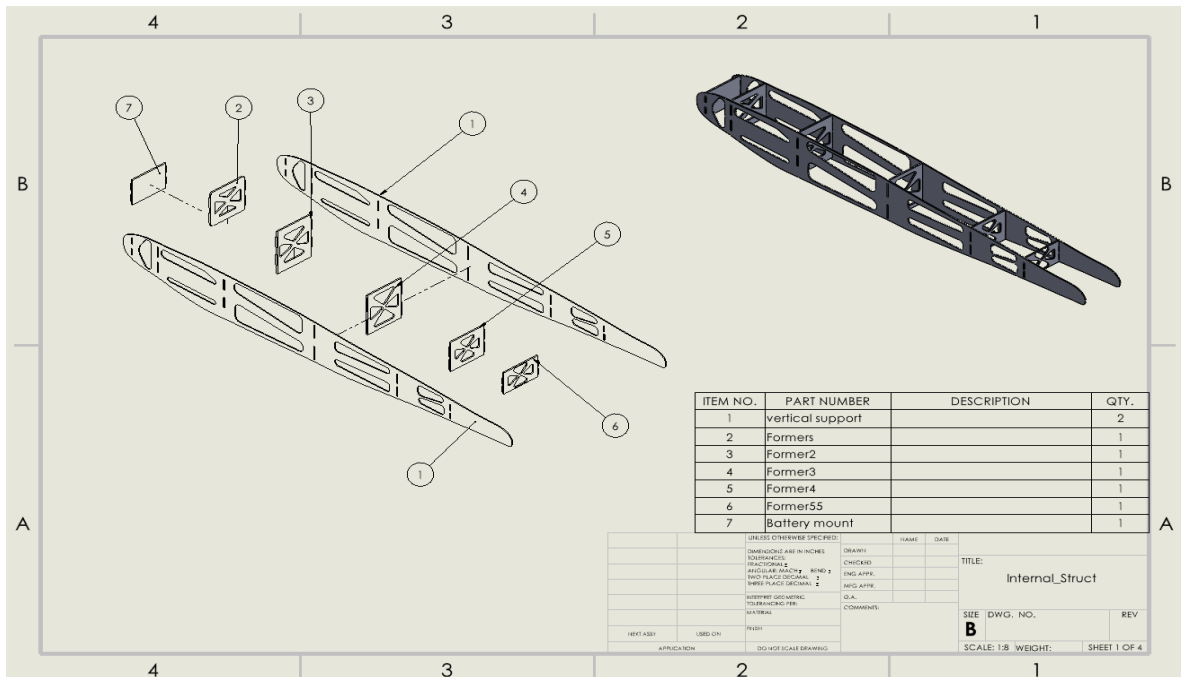


Figure 9.9-5 Internal Structure of the Fuselage, Sheet 1

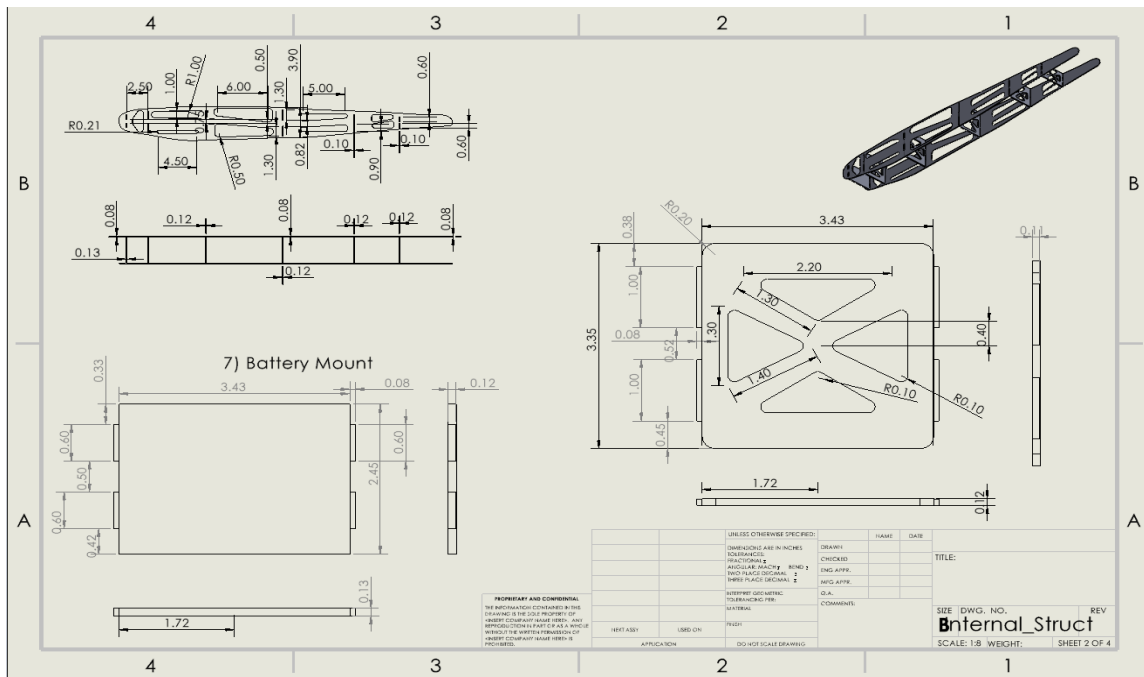


Figure 9-6 Internal Structure of the Fuselage, Sheet 2

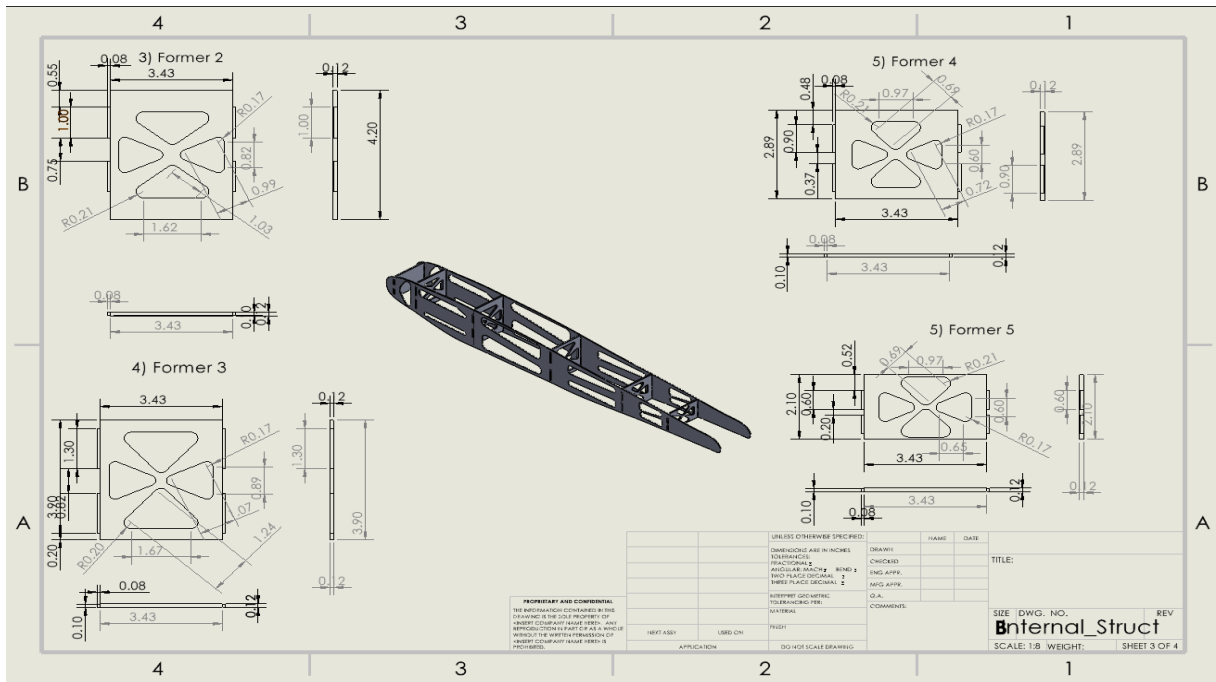


Figure 9-7 Internal Structure of the Fuselage, Sheet 3

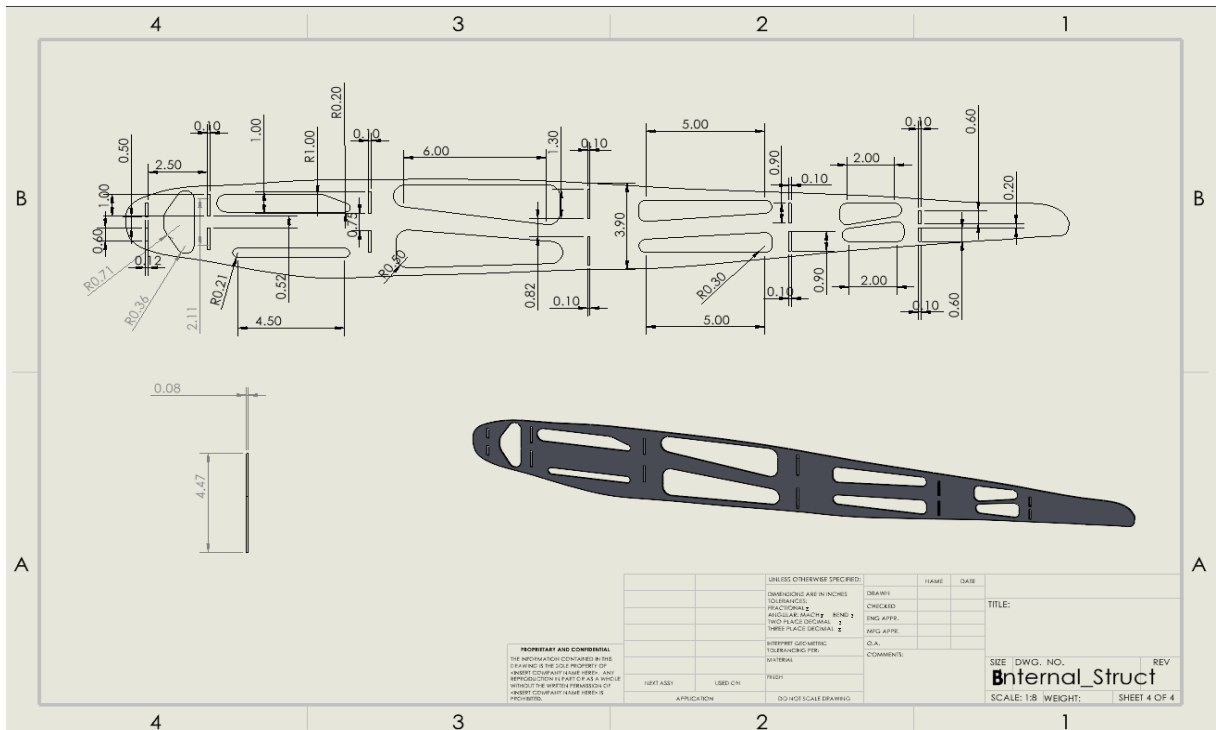


Figure 9-8 Internal Structure of the Fuselage, Sheet 4



10 COMPUTATIONAL FLUID DYNAMICS ANALYSIS [5]

10.1 WING CFD WITH DIFFERENT AILERON DEFLECTION

Computational Fluid Dynamics (CFD) is a powerful tool for predicting the aerodynamic characteristics of an RC plane, crucial for optimizing design, performance, and stability. By solving the Navier-Stokes equations, CFD allows the simulation of fluid flow around the plane, providing valuable data on pressure distribution, velocity fields, drag, and lift forces. In RC plane design, CFD can help analyze various components like wings, fuselage, stabilizers, and propellers individually and as a cohesive system. Here we had performed the CFD of the wing in three different domains at the same time with different ailerons deflection. In the top and bottom wing, the ailerons have positive and negative deflections respectively of 30 deg. While the middle domain has an aileron deflection of zero deg.

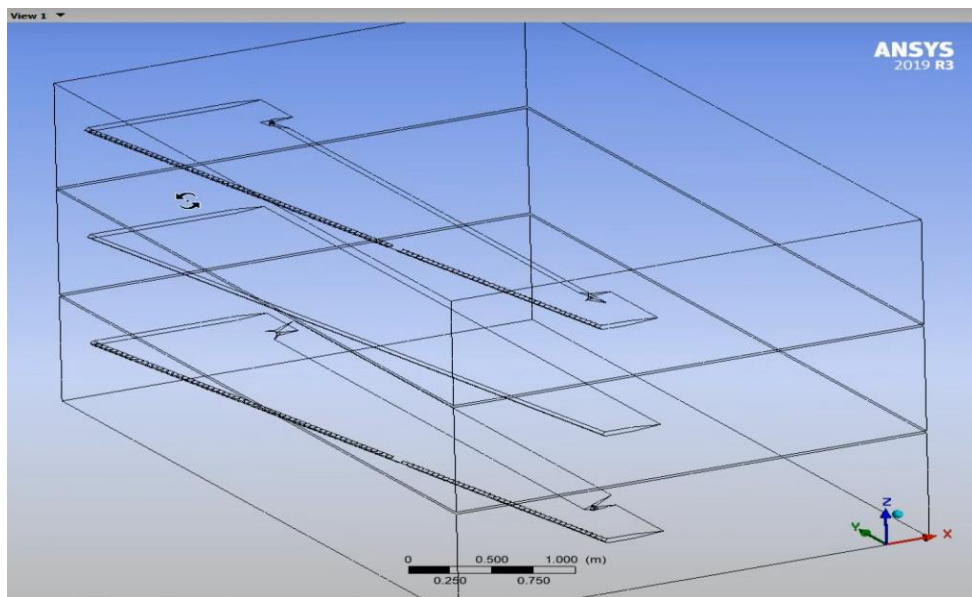


Figure 10-1 Wing Geometry



10.2 PRESSURE CONTOURS ON WING

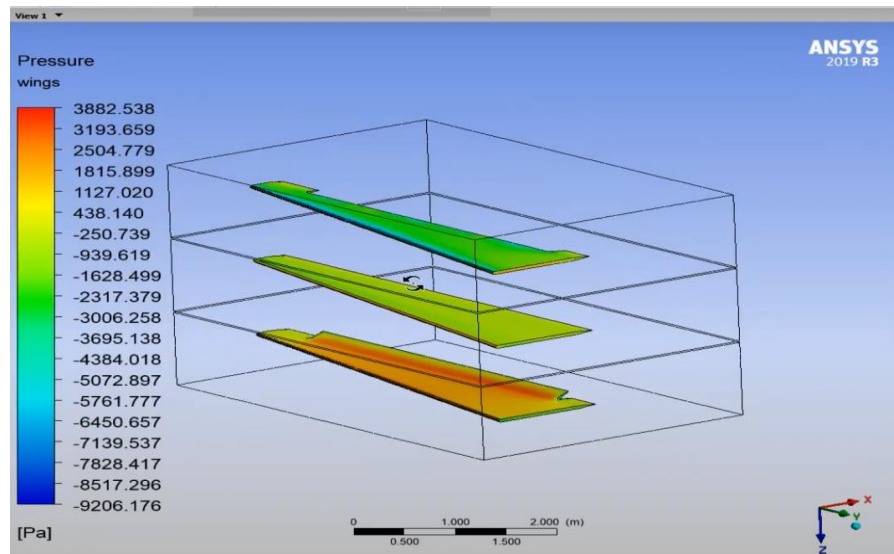


Figure 10-2 Pressure Contour on the Wings

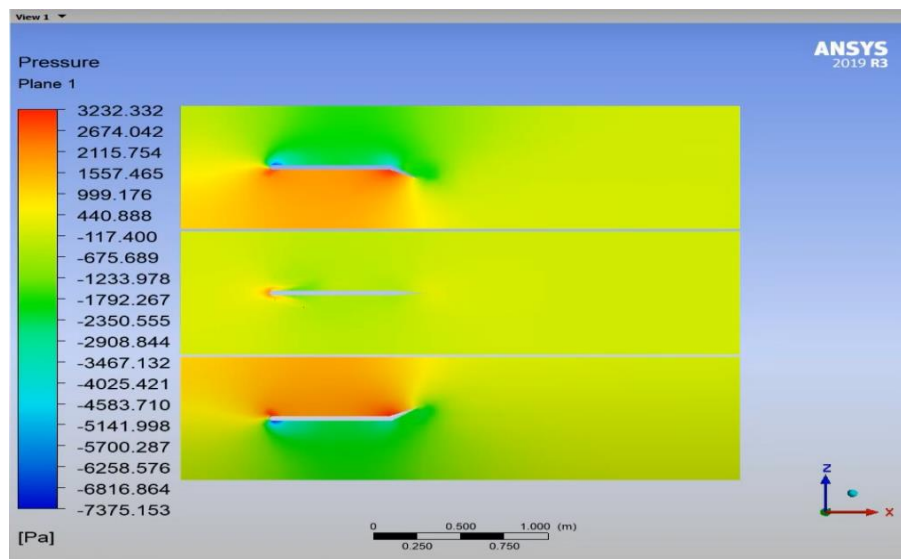


Figure 10-3 Pressure Contours on the Plane

10.2.1 Result

The above shown results are the local pressure results of the wing. It shows how the pressure is distributed for different ailerons deflection. When the aileron is at -30 degree the pressure is high at the bottom thus generating force from the bottom, similar is the case for the +30-degree deflection



10.3 VELOCITY CONTOURS ON WINGS



Figure 10-4 Velocity Streamline on the Wings

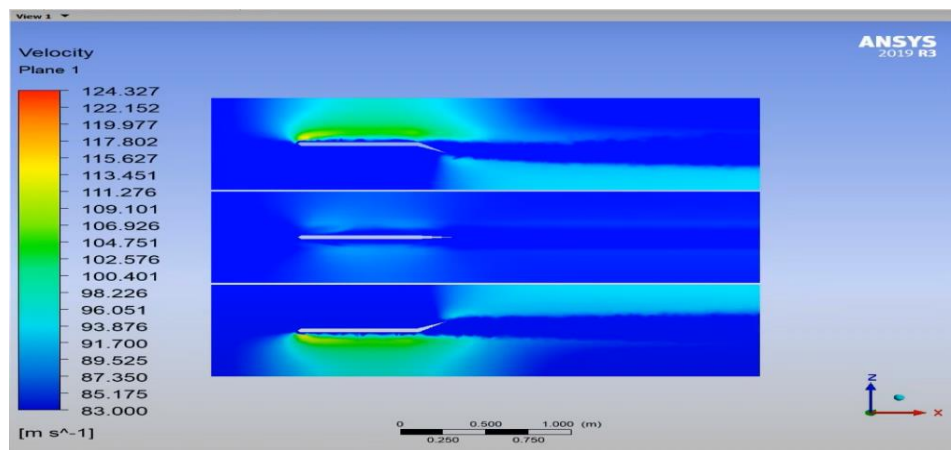


Figure 10-5 Velocity Contour on the Plane

10.3.1 Result

The above-mentioned figure shows the free stream velocity in three different scenarios. When the aileron deflection angle is zero the turbulence at the trailing is least but when the aileron is deflected turbulence occurs



10.4 HORIZONTAL TAIL CFD WITH DIFFERENT ELEVATOR DEFLECTION

10.4.1 Velocity Contours with Zero Elevator Deflection

The CFD was done on the Horizontal tail with no deflection of the elevator. The Lift and Drag plot were obtained along with the velocity contours. The velocity was shown in the streamline plot. The inlet free stream velocity was kept at 20 m/s.

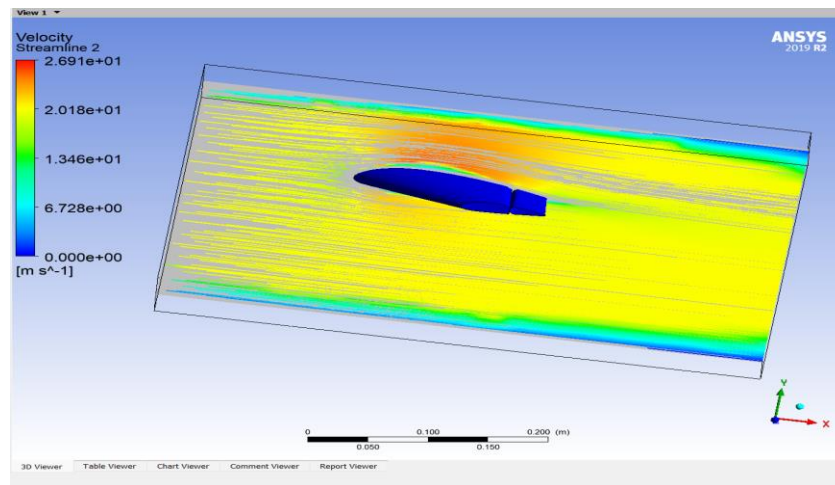


Figure 10-6 Velocity Streamline on Tail with zero Elevator Deflection

The streamline flow is shown below, very less turbulence is produced after the elevator because there is no deflection.

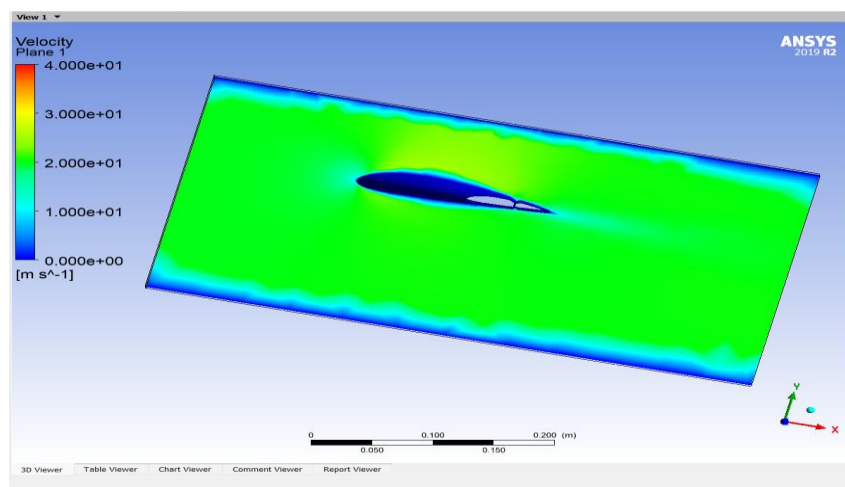


Figure 10-7 Velocity Contour on the Plane



10.5 PRESSURE CONTOURS WITH ZERO ELEVATOR DEFLECTION

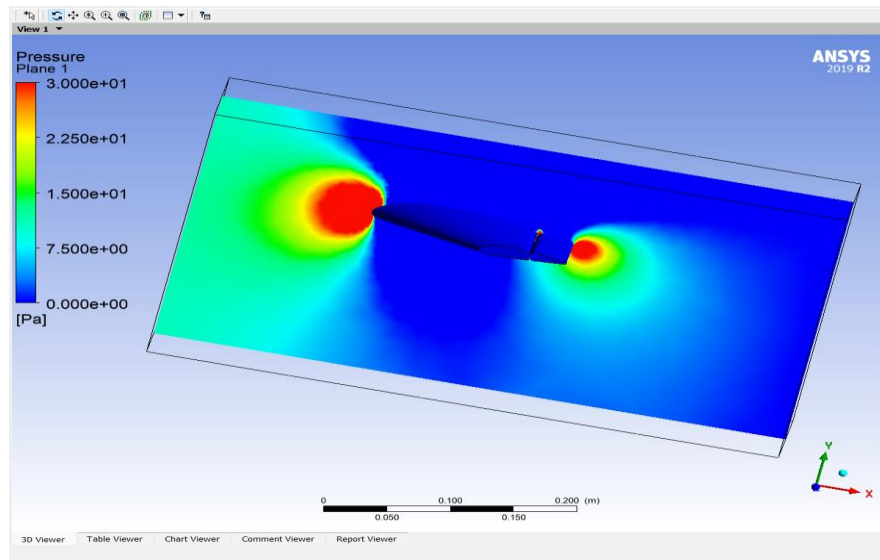


Figure 10-8 Pressure Contours on the Plane

Now we have the lift and drag plot for number of iterations.

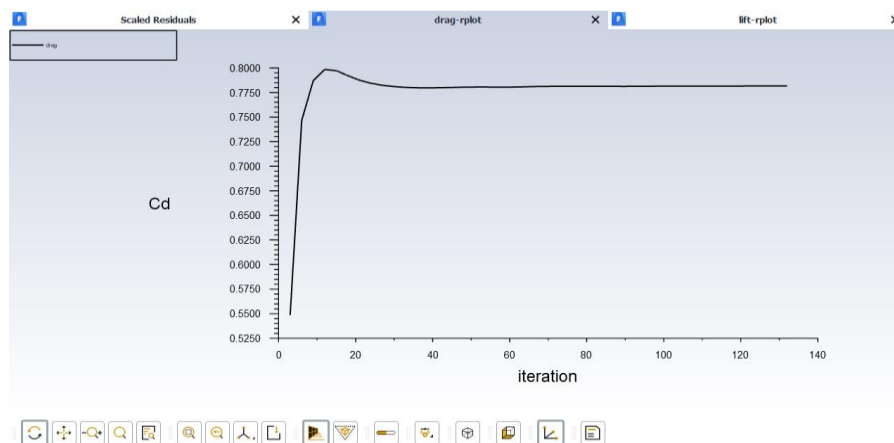


Figure 10-9 Coefficient of Drag Plot

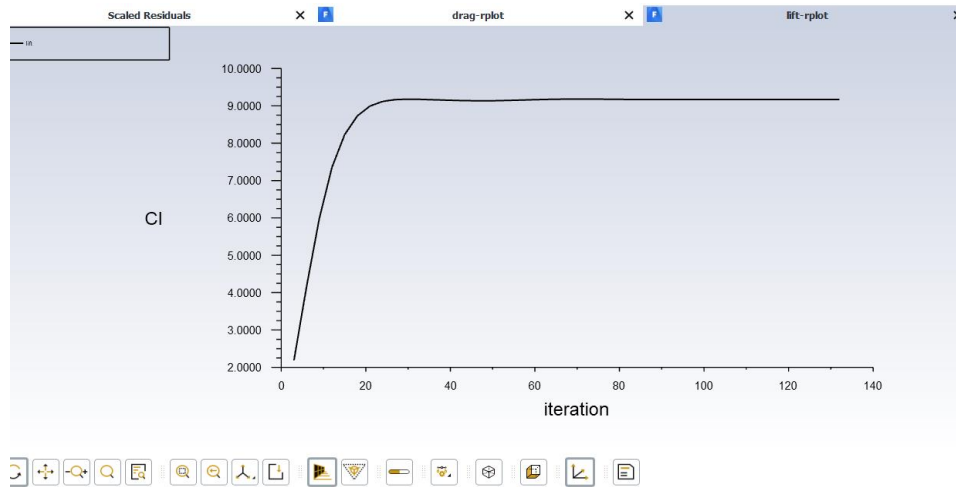


Figure 10-10 Coefficient of Lift Plot

10.6 VELOCITY CONTOURS WITH 30 DEGREE ELEVATOR DEFLECTION

The Elevator was given a positive angle of deflection of 30 degree and lift, drag plots were obtained along with Pressure and velocity contours. The free stream velocity was kept 20 m/s. The velocity contours are as follows.

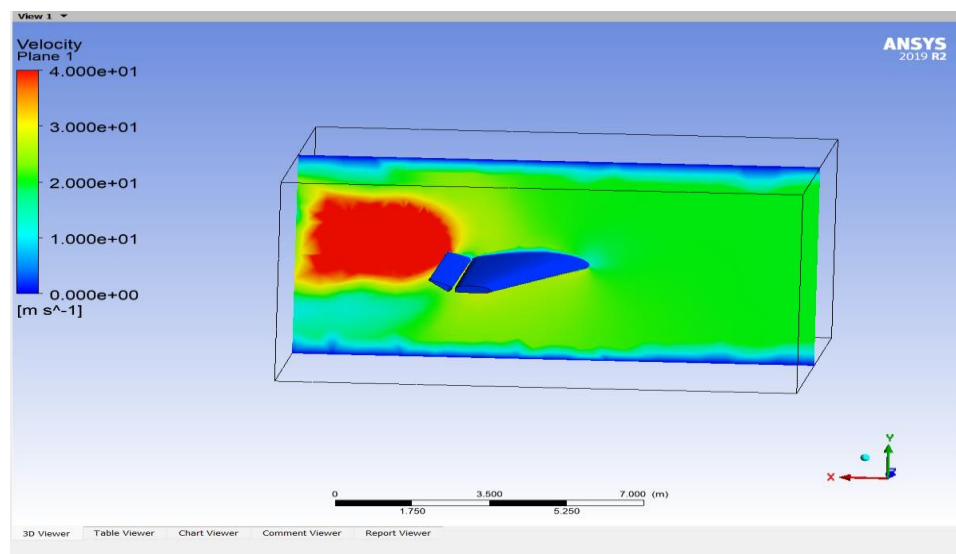


Figure 10-11 Velocity Contour with +30-degree Elevator Deflection

The streamline flow was also obtained which shows wake region greater turbulence occurs after the elevator.

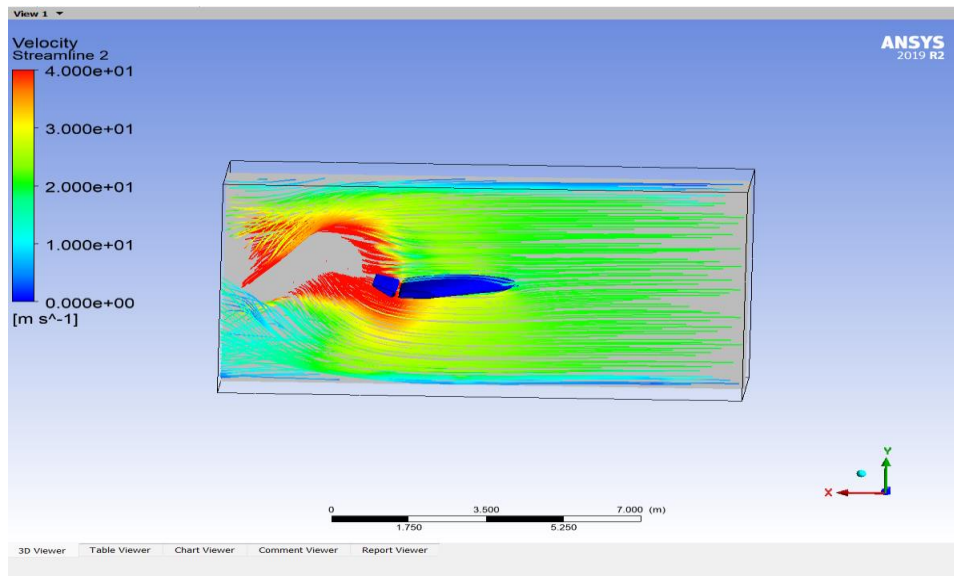


Figure 10-12 Velocity Streamline with +30-degree Elevator Deflection

We had also obtained the Pressure contours. The bottom of the horizontal tail has less pressure thus resulting in a downward force and producing the upward lift for the plane.

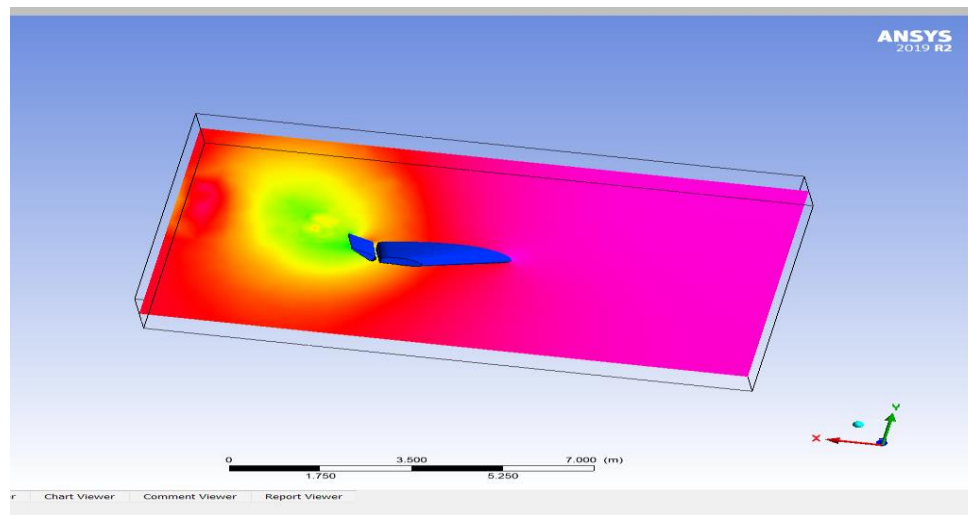


Figure 10-13 Pressure Contour on the Plane

The lift and drag plots for the number of iterations are as follows.

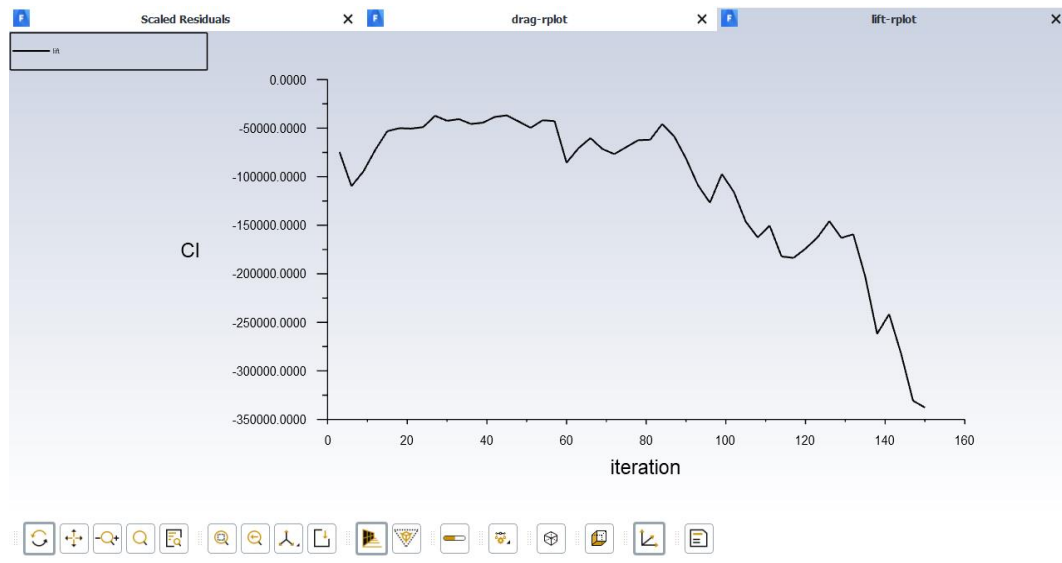


Figure 10-14 Coefficient of Lift



11 PERFORMANCE ANALYSIS [6]

11.1 STEADY LEVEL FLIGHT

An aircraft in an un-accelerated flight at a constant altitude is referred to as being in a state of steady level flight. In such a condition, the aircraft maintains a stable path with no change in altitude. When the aircraft is in steady level flight, the flight path angle is zero, and all forces and moments are balanced. The equations of force equilibrium can be succinctly summarized as follows: the thrust is equal to the drag, and the lift is equal to the weight.

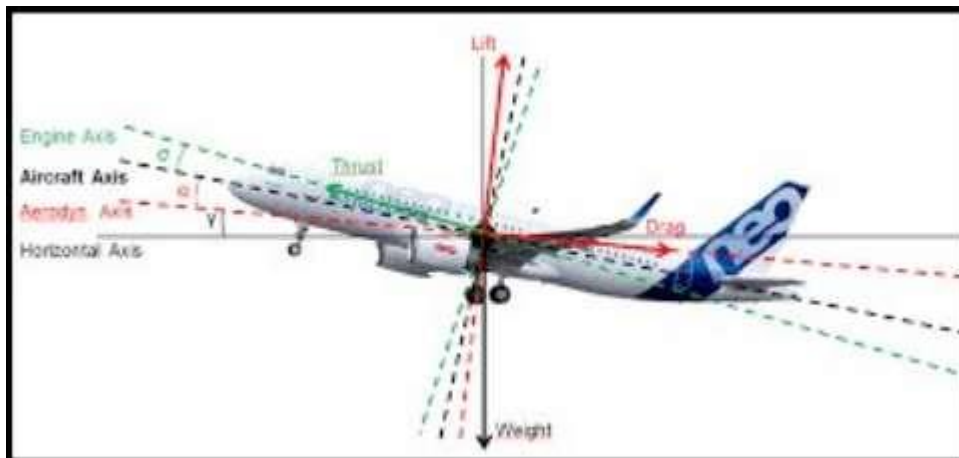


Figure 11-1 Steady Level Flight

Hence, the translational equations of motions reduce to: -

$$T = D = qS(C_{do} + C_L)$$

$$L = W = qSC_L$$

$$L = W = 21.56 \text{ N}$$

11.2 FLIGHT PERFORMANCE PARAMETER

The performance of any aircraft is dictated by its drag polar. So, we first find the drag polar and then evaluate important flight parameters for our airplane.



11.2.1 Parasite Drag

The Parasite drag was calculated using Open VSP model aerodynamic analysis. The drag caused by the landing gear has also been considered. A margin for fabrication error and the drag caused by it has also been included in the analysis. The result of the analysis is as follows:

$$C_{D0} = 0.03$$

Using this value of parasitic drag, induced drag was calculated with increasing angle of attack and the total drag was plotted against increasing Angle of Attack.

$$C_D = C_{D0} + C_{Di}$$

Since Open VSP does not account for boundary layer separation effects, these values just give a rough estimate and show the overall general trend, which is increasing.

11.2.2 Induced Drag Coefficient

The method for calculating induced drag coefficient is the one suggested in Raymer's Book of aircraft design. We know

$$C_D = C_{D0} + KC_l^2$$

We shall find the values of **K** and **C_{D0}**.

For finding K, we have

$$k = k_1 + k_2 + k_3$$

For subsonic Flight **K₂=0**,

For **K₃**

$$K_3 = \frac{1}{\pi e AR}$$



Assuming $e = 0.9$ (assumption is based on near elliptical lift distribution during lift analysis)

$$K_3 = 0.057$$

$$k_1 = \frac{k_3}{3}$$

$$k_1 = \frac{0.057}{3}$$

$$k_1 = 0.019$$

$$k = k_1 + k_3$$

$$k = 0.019 + 0.057 = 0.076$$

$$\mathbf{k = 0.076}$$

11.2.3 Drag Polar

After evaluating both the values for K and C_{d0} the drag polar is given by

$$C_d = C_{D_0} + KC_L^2$$

$$C_d = 0.03 + 0.076C_L^2$$

Using this drag polar we calculate some of the important airplane properties as follows:

11.3 MINIMUM THRUST REQUIRED IN LEVEL FLIGHT

The term "Thrust Required" denotes the minimal thrust necessary to counteract the drag opposing the aircraft's forward movement at a particular speed and altitude. To quantify this minimum thrust requirement for level flight, we can employ the expression presented on page 458 of Raymer's 1992 publication, as provided below:

$$T_R = \sqrt{4C_{d0}K * W}$$

$$W = 21.56N$$



$$S = 0.314m^2$$

$$C_{do} = 0.03$$

$$K = 0.076$$

$$\rho = 1.225 \frac{kg}{m^3}$$

$$T_R = \sqrt{4(0.03)(0.076) * 21.56}$$

$$\mathbf{T_R = 0.443 N}$$

Velocity at which thrust required is minimum is:

$$V_{TR-min} = \sqrt{\frac{2}{\rho} * \sqrt{\frac{K}{C_{do}}} * \frac{W}{S}}$$

$$V_{TR-min} = \sqrt{\frac{2}{1.225} * \sqrt{\frac{0.076}{0.03}} * \frac{21.56}{0.314}}$$

$$\mathbf{V_{TR-min} = 4.265 \frac{m}{s}}$$

MINIMUM THRUST

The method for calculating the minimum thrust required is by using an iterative method.

The main formulas that are involved are as follows

$$C_l = \frac{2W}{\rho * V^2 * S}$$

$$T_r = \frac{\rho * V^2 * S * C_d}{2}$$

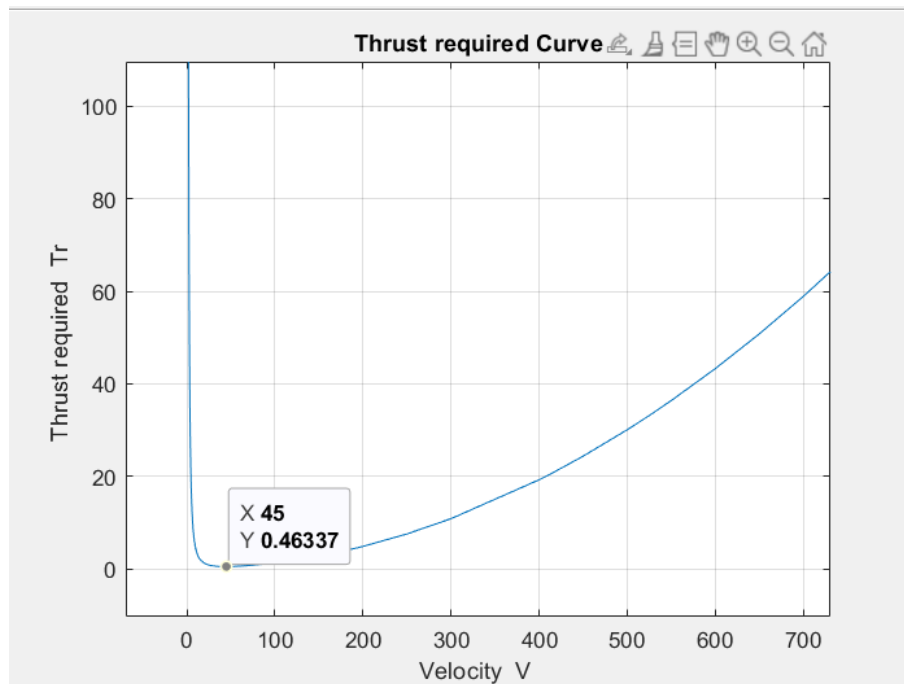


Figure 11-2 Thrust Required Curve

From this graph we can see that,

$$V_{Tr(min)} = 45 \text{ ft/s}$$

And the thrust required minimum is,

$$Tr_{(min)} = 0.4633 \text{ N}$$

COEFFICIENT OF LIFT AT MINIMUM THRUST

$$C_{L-min} = \frac{\sqrt{C_{do}}}{K}$$

$$C_{L-min} = \frac{\sqrt{0.03}}{0.076}$$

$$C_{L-min} = 2.279$$

11.4 AERODYNAMIC RELATIONS

Variation of L/D with the change in velocity of aircraft is given as:



$$\frac{L}{D} = \left(\left(\frac{\rho V^2 C_{do} S}{2W} \right) + \left(\frac{2KW}{\rho V^2 S} \right) \right)$$

$$\frac{L}{D} = \left(\left(\frac{(1.225)(13)^2(0.03)}{2(7)} \right) + \left(\frac{2(0.076)}{(1.225)(13)^2 * 7} \right) \right)^{-1}$$

$$\frac{L}{D} = \mathbf{8.00}$$

$\left(\frac{C_L^{3/2}}{C_D} \right)_{\max}$ is calculated as;

$$\left(\frac{C_L^{3/2}}{C_D} \right) = \frac{1}{4} \left(\frac{3}{K C_{do}^3} \right)^{\frac{3}{4}}$$

$$\left(\frac{C_L^{3/2}}{C_D} \right) = \frac{1}{4} \left(\frac{3}{(0.076)(0.03^3)} \right)^{\frac{3}{4}}$$

$$\left(\frac{C_L^{3/2}}{C_D} \right)_{\max} = \mathbf{9.46}$$

The corresponding velocity is as below

$$V_{\left(\frac{C_L^{3/2}}{C_D} \right)} = \sqrt{\frac{2}{\rho} * \sqrt{\frac{K}{3 C_{do}}} * \frac{W}{S}}$$

$$V_{\left(\frac{C_L^{3/2}}{C_D} \right)} = \sqrt{\frac{2}{1.225} * \sqrt{\frac{0.76}{3 * 0.03}} * 7}$$

$$V_{\left(\frac{C_L^{3/2}}{C_D} \right)} = \mathbf{3.24 \text{ m/s}}$$



$\left(\frac{C_L}{C_D}\right)_{\max}$ is calculated as

$$\left(\frac{C_L}{C_D}\right)_{\max} = \sqrt{\frac{1}{4C_{d0}K}}$$

$$\left(\frac{C_L}{C_D}\right)_{\max} = \sqrt{\frac{1}{4 * 0.03 * 0.076}}$$

$$\left(\frac{C_L}{C_D}\right)_{\max} = \mathbf{10.47}$$

The corresponding velocity is as below

$$V_{\frac{C_L}{C_D}} = \sqrt{\frac{2}{\rho} * \sqrt{\frac{K}{C_{d0}} * \frac{W}{S}}}$$

$$V_{\frac{C_L}{C_D}} = \sqrt{\frac{2}{1.225} * \sqrt{\frac{0.076}{0.03} * 7}}$$

$$V_{\frac{C_L}{C_D}} = \mathbf{4.265 \text{ m/s}}$$

$\left(\frac{C_L^{1/2}}{C_D}\right)_{\max}$ is calculated as

$$\left(\frac{C_L^{1/2}}{C_D}\right)_{\max} = \frac{3}{4} \left(\frac{1}{3KC_{d0}^3}\right)^{\frac{1}{4}}$$

$$\left(\frac{C_L^{1/2}}{C_D}\right)_{\max} = \frac{3}{4} \left(\frac{1}{3 * 0.076 * 0.03^3}\right)^{\frac{1}{4}}$$

$$\left(\frac{C_L^{1/2}}{C_D}\right)_{\max} = \mathbf{15.06}$$

The corresponding velocity is as below



$$V_{\frac{1}{\frac{C_L^2}{C_D}}} = \sqrt{\frac{2}{\rho} * \sqrt{\frac{3K}{C_{d0}}} * \frac{W}{S}}$$

$$V_{\frac{1}{\frac{C_L^2}{C_D}}} = \sqrt{\frac{2}{1.225} * \sqrt{\frac{3 * 0.076}{0.03}} * 7}$$

$$V_{\frac{1}{\frac{C_L^2}{C_D}}} = 5.613 \text{ m/s}$$

We discovered that the velocity ratios were nearly identical to those predicted by Anderson, and they were repeated as follows.

$$V_{\frac{3}{\frac{C_L^2}{C_D}}} : V_{\frac{C_L}{C_D}} : V_{\frac{1}{\frac{C_L^2}{C_D}}} = 3.24 : 4.265 : 5.613$$

11.5 MINIMUM POWER REQUIRED IN LEVEL FLIGHT

Power is usually defined as the product of force and velocity but since the force in question equals the amount of drag under steady level conditions, using the expression featured on page 458 (Raymer, 1992) given below

$$P_R = \sqrt{\frac{2W^3}{PS \left(\frac{C_L^3}{C_D} \right)}}$$

$$P_R = \sqrt{\frac{2(21.56)^3}{(1.225)(0.314)(9.46^2)}}$$

$$P_R = 21.13 \text{ W}$$

To determine the minimum power required, we should consider a velocity equal to 0.76 times the velocity required for minimum thrust. This approach is in accordance with the expression on page 457 of Raymer's 1992 publication, which is as follows



$$V_{PR-min} = V_{\frac{C_L^2}{C_D}} = \sqrt{\frac{2}{\rho} * \sqrt{\frac{K}{3C_{d0}}} * \frac{W}{S}}$$

$$V_{PR-min} = V_{\frac{C_L^2}{C_D}} = \sqrt{\frac{2}{1.225} * \sqrt{\frac{0.076}{3 * 0.03}} * 7}$$

$$V_{PR-min} = 3.24 \text{ m/s}$$

We get the same result using both the graphical and analytical approaches: the minimum power needed. One point to note here is that the minimum power needed occurs at a velocity lower than the minimum thrust required.

$$V_{PR-min} < V_{TR-min}$$

11.6 RATE OF CLIMB

The rate of climb 'ROC' of an aircraft is simply its vertical speed at any given time during flight, but it is used specifically to assess the climb performance of the aircraft. The opposite of the climb rate is the sink or descent rate which will be investigated later in this section. Although the ROC of an aircraft varies significantly with altitude, the analysis is only being performed at cruise height since it remains at that altitude throughout most of its flight. Generally, an aircraft is designed to have an optimum performance at the cruise altitude as well, since this provides a reference point. To obtain an analytical value for the maximum rate of climb of the aircraft 'ROCmax', the expression on page 276 (Anderson, 1999) featured below can be used:

$$ROC_{max} = \left[\frac{(\frac{W}{S})Z}{3\rho C_{D0}} \right]^{\frac{1}{2}} * \left(\frac{T}{W} \right)^{\frac{3}{2}} * \left[1 - \frac{Z}{6} - \frac{3}{2(\frac{T}{W})^2(\frac{L}{D})^2Z} \right]$$

Where



$$Z = 1 + \sqrt{1 + \frac{3}{2\left(\frac{T}{W}\right)^2\left(\frac{L}{D}\right)_{max}^2}}$$

$$S = 0.314 \text{ m}^2$$

$$C_{do} = 0.03$$

$$K = 0.076$$

$$\rho = 1.225 \frac{\text{kg}}{\text{m}^3}$$

$$\frac{L}{D} = 8$$

$$Z = 2$$

$$\mathbf{ROC = 9 \text{ ft/sec}}$$

The velocity at this rate of climb can also be evaluated using the expression featured on page 275 (Anderson, 1999), given below:

$$V(R/C)_{max} = \left\{ \left[\frac{\left(\frac{W}{S}\right)\left(\frac{T}{W}\right)}{3\rho C_{D_0}} \right] \left(1 + \sqrt{1 + \frac{3}{2\left(\frac{T}{W}\right)^2\left(\frac{L}{D}\right)^2}} \right) \right\}^{\frac{1}{2}}$$

$$\mathbf{V(R/C)_{max} = 12 \text{ ft/sec}}$$

11.7 ENDURANCE

Endurance calculations are also performed for unaccelerated flight using the following relationship:

$$\text{Endurance(hours)} = \frac{\text{BatteryCapacity(Wh)}}{\text{AverageCurrentDraw(A)}}$$

$$E_{max} = \frac{2.2}{12}$$

$$\mathbf{E_{max} = 11min}$$



11.8 MAXIMUM RANGE

Range is the total distance (measured with respect to the ground) traversed by an airplane on one load of fuel. Range for a battery powered RC Plane is given by

Range (miles/kilometers) = Endurance (hours) * Average Speed (miles/kilometers per hour)

$$R_{\max} = 0.183 \times 50$$

$$R_{\max} = 9.15 \text{ km}$$

11.9 GLIDE PERFORMANCE

Gliding (Unpowered flight) is the ability to stay airborne for a period without using any power source. Two parameters are important for unpowered flight.

Minimum glide angle

Maximum range

MINIMUM GLIDE ANGLE

For minimum glide angle $\theta_{\min}$,

$$\tan \theta_{\min} = \sqrt{4C_{d0}k}$$

$$\tan \theta_{\min} = 0.095$$

$$\theta_{\min} = 5.455 \text{ deg}$$

MAXIMUM GLIDE RANGE

$$\frac{h}{R} = \tan \theta_{\min}$$

$$R = \frac{h}{\tan \theta_{\min}}$$

$$R = \frac{100}{0.095}$$

$$R = 1052.63 \text{ ft}$$



The velocity at minimum glide angle at sea level will be,

$$V_{(L/D)\max} = \left(\frac{2W\sqrt{K/C_{do}}}{\rho S} \right)^{\frac{1}{2}}$$

$$V_{(L/D)\max} = \left(\frac{(0.798)2\sqrt{0.058/0.016}}{(0.00237)} \right)^{\frac{1}{2}}$$

$$V_{(L/D)\max} = 43.8 \text{ ft/s}$$

11.10 ECALC PERFORMANCE

11.10.1 Propulsive Performance

The propulsive performance is estimated using the ECalc propellor performance calculator using the closest possible values as our components. It provides for an estimation of the flight time and possible thrust by weight ratio. The Flight time calculated is for 85% power discharge.

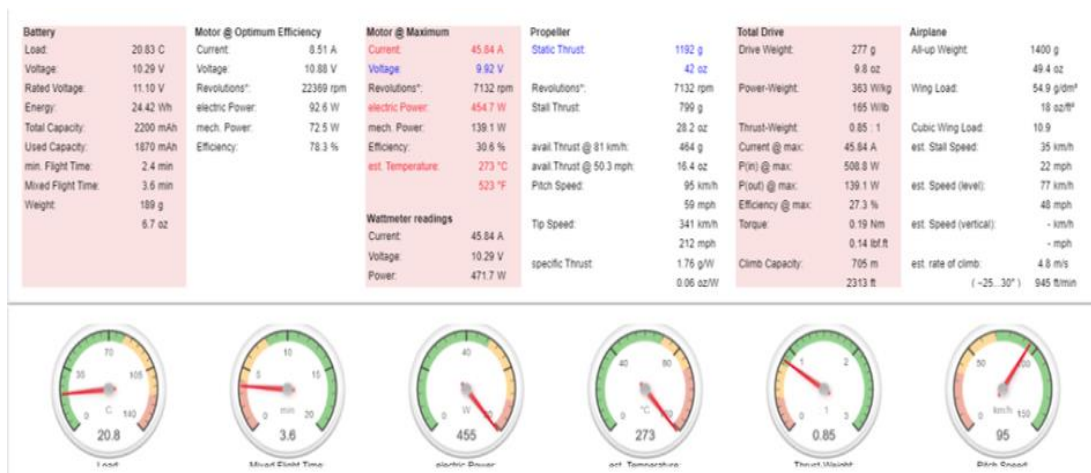


Figure 11-3 Propulsive Performance Using ECalc

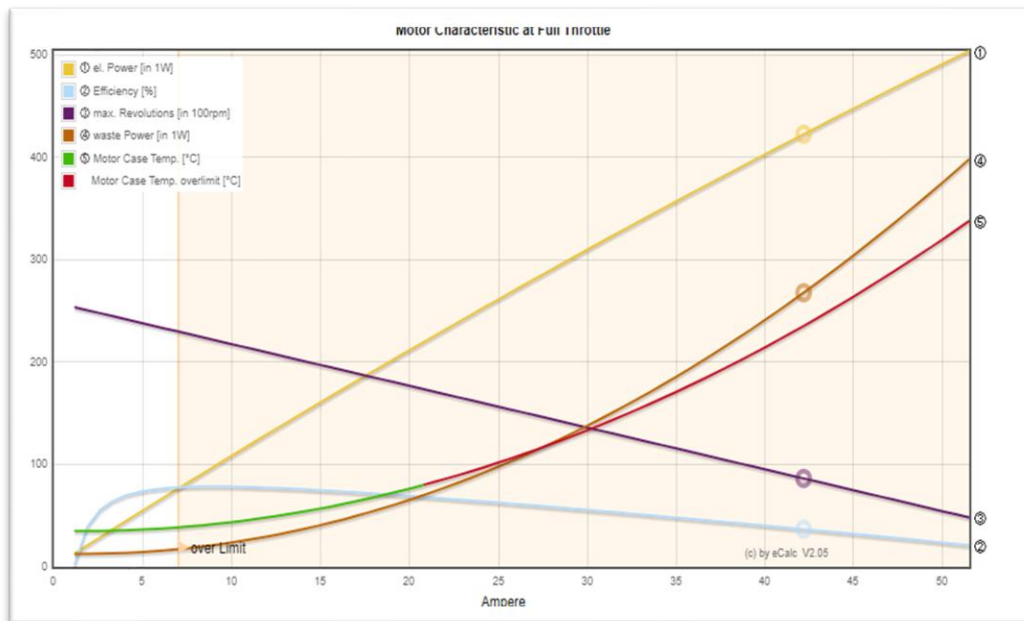


Figure 11-4 Propulsion Performance Graph



12 STABILITY ANALYSIS [7]

GLIDE PERFORMANCE

In the Aerospace sector, stability analysis is essential for making sure that aircraft can continue to fly under control settings in a variety of situations. It examines how changes in forces, such as lift and drag, affect an aircraft's ability to return to equilibrium after disturbances. Engineers can improve safety features and find design defects by examining static and dynamic stability. We used DATCOM to calculate the coefficient of moments about the three axes. These results give the moments coefficient of C_{m_α} , C_{n_β} , C_{L_α} etc.

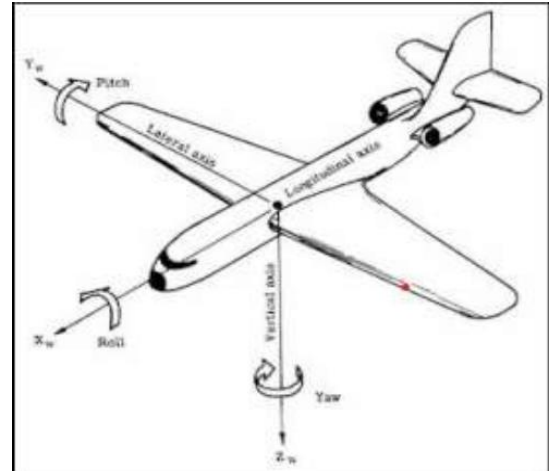


Figure 12-1 Stability Axis of an Aircraft

12.1 STABILITY DERIVATIVES OVERVIEW

The application of DATCOM enabled computation of key stability derivatives: C_{n_β} (yaw stability), C_{L_β} (roll stability), C_{m_α} (pitch stability), C_{L_α} (rolling moment coefficient due to angle of attack), C_{y_β} (side force coefficient due to sideslip angle) etc. Some of the stability derivatives are discussed below,

12.1.1 Directional Stability Derivative

C_{n_β} : The yawing moment due to sideslip is represented by the stability derivative C_{n_β} . It measures the amount of yawing moment (turning force around the vertical axis) produced when the aircraft encounters a sideslip angle (β), which occurs when the wind or other disturbances cause the nose of the aircraft to point slightly away from its forward path.



12.1.2 Lateral Stability Derivative

$C_{L\beta}$: It is the rolling moment caused by sideslips. When the aircraft experiences a sideslip, it shows the amount of roll, or rotation along the longitudinal axis, that is produced.

$C_{y\beta}$: It represents the side force generated when there's a sideslip. A positive $C_{y\beta}$ creates a restoring force that pushes the aircraft back in line with its intended path, aiding in directional stability. This helps maintain course during crosswinds or yaw disturbances.

12.1.3 Longitudinal Stability Derivative

$C_{m\alpha}$: A change in the angle of attack (α) produces a pitching moment, which is indicated by $C_{m\alpha}$ (Pitching Moment owing to Angle of Attack). With a negative $C_{m\alpha}$, an increase in angle of attack results in a nose-down moment that helps the airplane recover to its initial attitude following a disturbance, stabilizing the pitch.

C_{m_q} (**Pitching Moment Coefficient due to Pitch Rate**): Pitching Moment owing to Pitch Rate, or C_{m_q} is the pitching moment produced by the pitch change rate (q). A negative C_{m_q} dampens pitch, reducing sudden changes in the aircraft's pitch angle and assisting in the reduction of pitch oscillations and dynamic stability.

12.1.4 Roll Stability Derivative

$C_{L\alpha}$: The rolling moment that results from a change in the angle of attack (α) is represented by the symbol $C_{L\alpha}$. Roll stability and control may be impacted by a positive $C_{L\alpha}$, which can lead the aircraft to roll in the direction of an increased angle of attack.

C_{l_p} : It is a measure of the rolling moment produced by the roll rate (p) of the aircraft. Roll damping is provided by a negative C_{l_p} , which reduces roll oscillations and stabilizes the roll motion by countering fast rolling movements.



12.1.5 Yaw Stability Derivative

C_{n_r} : It is the yawing moment caused by the yaw rate (r). Yaw damping, which stabilizes the aircraft's directional motion by preventing abrupt yawing and smoothing out oscillations, is facilitated by a negative C_{n_r} .

12.2 DATCOM MODEL

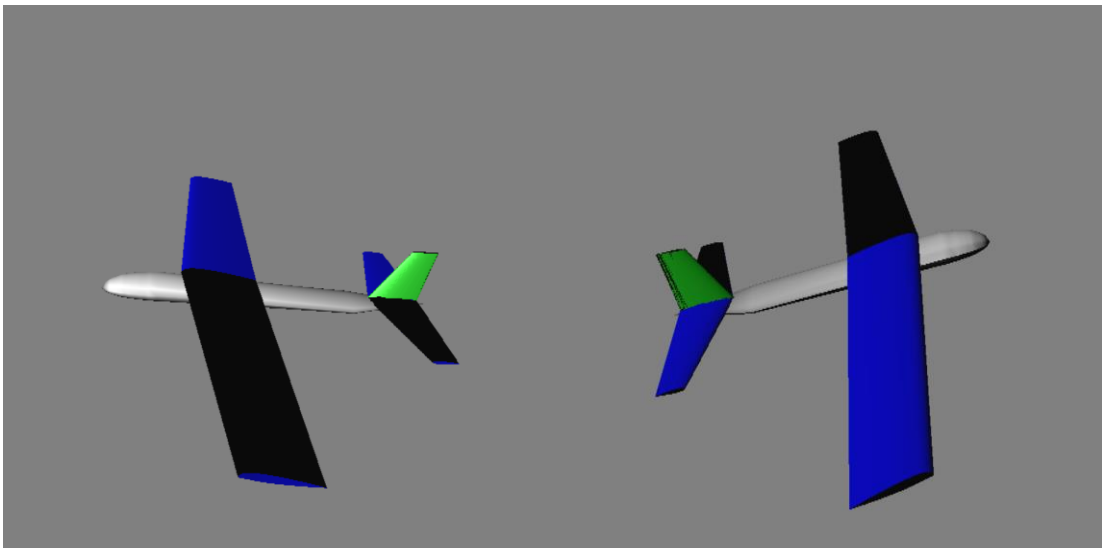


Figure 12-2 Side Views of Our Plane

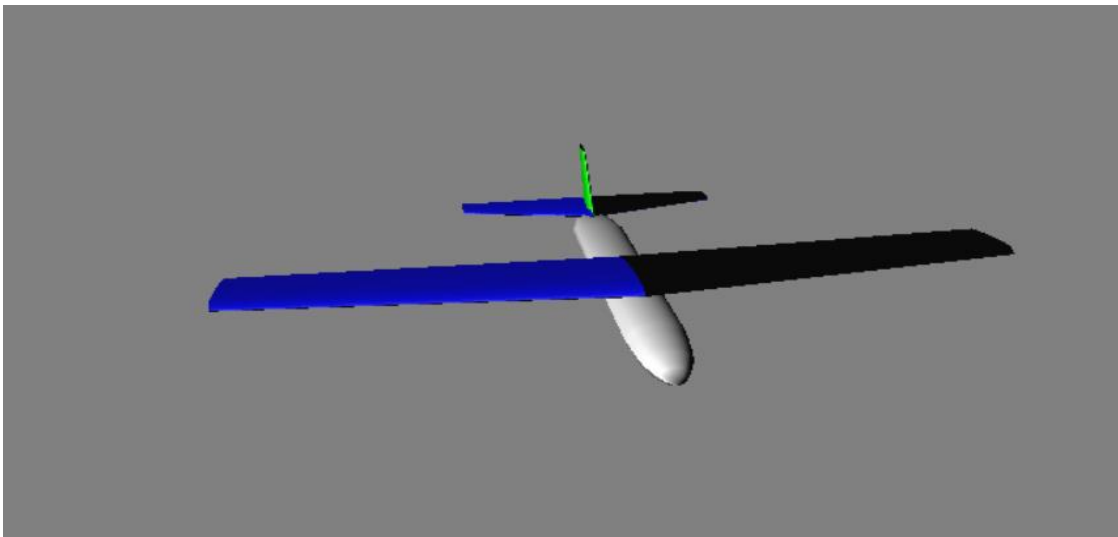


Figure 12-3 Datcom Model of Our Plane



12.3 ANALYSIS OF RESULT

We obtained several stability derivatives and interpreted each one as follows:

FLIGHT CONDITIONS						REFERENCE DIMENSIONS					
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LENGTH	MOMENT	REF. CENTER		
	FT	FT/SEC	LB/FT**2	DEG R	1/FT	FT**2	FT	FT	FT		
0	.400	.00	446.53	2.1162E+03	518.670	2.8273E+06	2719.600	51.160	319.870		
0								75.000	11.400		
0	-----DERIVATIVE (PER DEGREE)-----										
0	ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA		
								CYB	CNB		
									CLB		
-16.0	.428	-6.824	.8813	-6.677	-1.469	-.132	5.551E-01	-7.228E-02	-8.345E-03		
-8.0	.086	-2.441	.3292	-2.430	-.254	-.135	5.405E-01	-6.575E-02			
-6.0	.053	-1.364	.1993	-1.362	-.090	-.146	5.276E-01	-6.315E-02			
-4.0	.039	-.331	.0766	-.333	.016	-.230	5.113E-01	-6.061E-02			
-2.0	.043	.681	-.0431	.679	.067	-.063	5.169E-01	-6.165E-02			
.0	.066	1.736	-.1700	1.736	.066	-.098	5.392E-01	-6.509E-02			
2.0	.111	2.838	-.3035	2.840	.012	-.107	5.603E-01	-6.869E-02			
4.0	.180	3.978	-.4447	3.981	-.098	-.112	5.717E-01	-7.311E-02			
6.0	.387	6.278	-.7571	6.271	-.491	-.121	5.142E-01	-7.231E-02			
8.0	.442	6.777	-.8279	6.763	-.624	-.122	4.715E-01	-6.961E-02			
10.0	.496	7.221	-.8963	7.198	-.765	-.125	4.209E-01	-6.659E-02			
12.0	.599	7.970	-1.0224	7.921	-1.071	-.129	3.206E-01	-5.955E-02			
14.0	.683	8.504	NA	8.417	-1.394	NA	1.049E-01	NA			
16.0	.678	8.390	NA	8.252	-1.661	NA	-5.695E-01	NA			
18.0	.433	6.226	NA	6.055	-1.512	NA	-9.149E-01	NA			
19.0	.393	5.394	NA	5.228	-1.385	NA	-7.915E-01	NA			
20.0	.361	4.643	NA	4.486	-1.249	NA	-7.521E-01	NA			
21.0	.344	3.890	NA	3.755	-1.073	NA	-6.359E-01	NA			
22.0	.352	3.371	NA	3.258	-.936	NA	-5.104E-01	NA			
24.0	.389	2.385	NA	2.337	-.615	NA	-4.753E-01	NA			

0*** NA PRINTED WHEN METHOD NOT APPLICABLE

1 AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM

DYNAMIC DERIVATIVES

Figure 12-4 Stability Derivatives

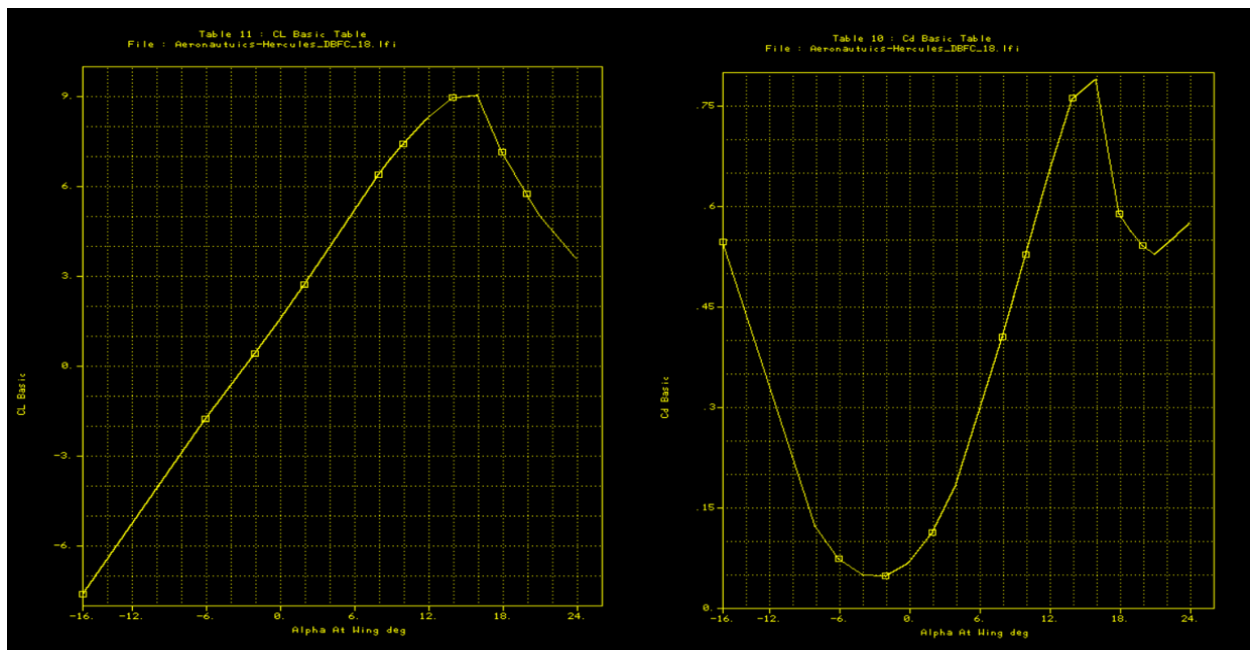


Figure 46 Lateral Stability (C_{l_p})



In general, a negative value of $C_{l\beta}$ typically indicates stable rolling motion. When we analyze the graph below then we can see that we have negative values of $C_{l\beta}$ which suggests that the aircraft will produce a rolling moment that tends to roll the aircraft in the direction of the sideslip when it encounters a sideslip, which is a relative wind direction that is not parallel to the aircraft's forward motion. The airplane is stabilized laterally, and the sideslip is counteracted by this opposing roll. Essentially, the aircraft will roll in the opposite direction of the sideslip, allowing it to return to level flight and preserve lateral stability.

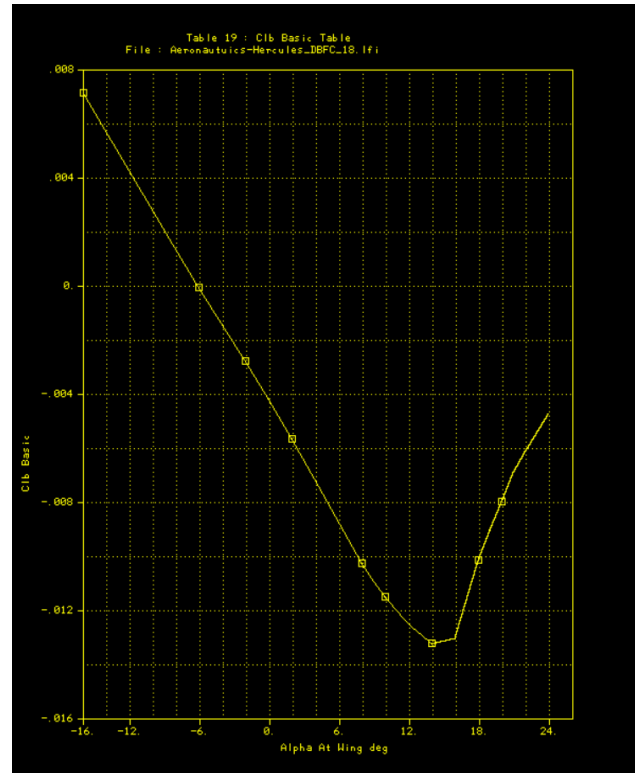


Figure 12-5 Graph of Cl-beta

12.4 PITCH STABILITY ($C_{m\alpha}$)

When we analyze the graph, then we can see that the pitching moment coefficient C_m decreases almost linearly as the angle of attack (α) increases. A negative $C_{m\alpha}$ ensures that the model develops static pitch stability, which acts against the development of pitch instability.

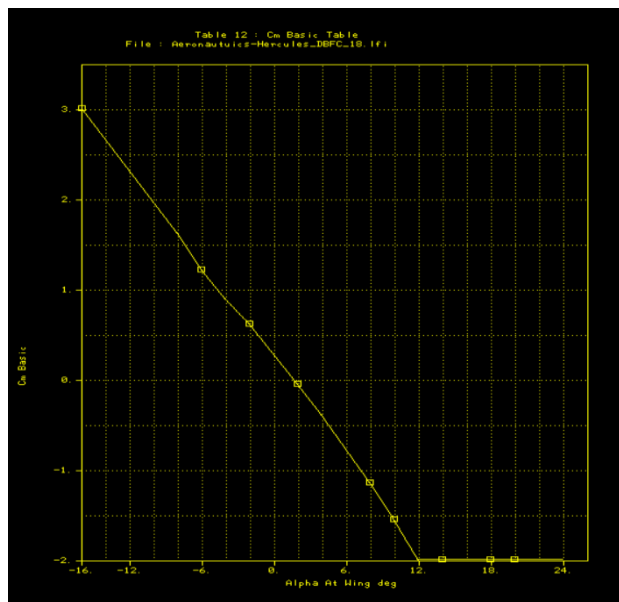


Figure 12-6 Graph of Cm alpha



12.5 ROLL RESPONSE (C_{l_α})

As we can see from the graph below that the C_{l_α} becomes negative as the angle of attack increases this means that the rolling moment produced by the aircraft is in the opposite direction of the increased angle of attack. In other words, the aircraft will produce a rolling moment that cancels out an increase in the angle of attack on one wing, so enhancing roll stability.

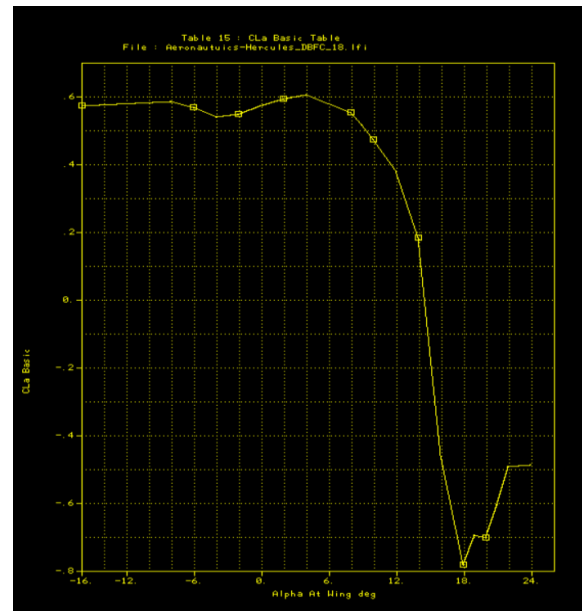


Figure 12-7 Graph of Cl alpha

12.6 DIRECTIONAL STABILITY ($C_{n\beta}$)

When we analyze the graph then we can see that it shows a positive $C_{n\beta}$, indicating strong directional stability. This confirms that the model will effectively correct itself in the yaw axis when disturbed. Directional stability is crucial for maintaining the aircraft's heading. Thus, the model's design ensures reliable yaw correction.

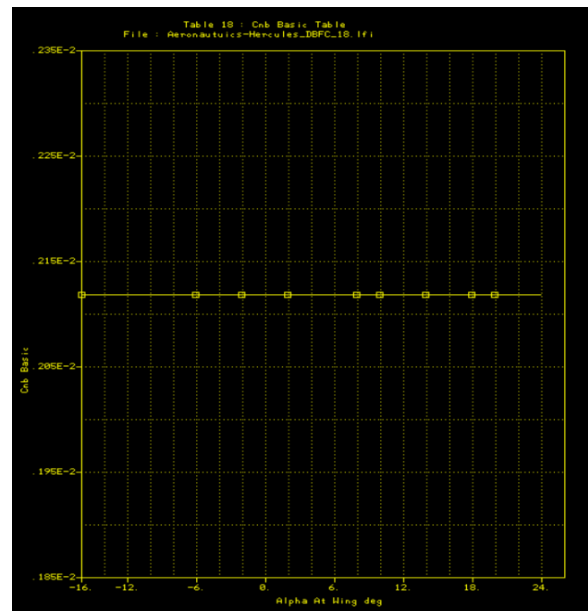


Figure 12-8 Graph of Cn Beta



12.7 STATIC VS DYNAMIC STABILITY

12.7.1 Static Stability

This is the first response of the aircraft to change of position from the equilibrium point, or in other words the first oscillation. According to the positive static derivatives ($C_{n\beta}$, C_{ma} and $C_{y\beta}$), it ensures that its original state will be provided by nature.

12.7.2 Dynamic Stability

Dynamic Stability refers to the period required for oscillations to die out. Some of the dynamic stability parameters that we calculated showed that the plane has negative damping about the roll, yaw, pitch axes making the plane easier to control as it flies.

12.7.3 Conclusion

According to our DATCOM analysis, the stability of the model is proven along all three axes to meet the competition needs. The positive static and dynamic stability characteristics demonstrate that the aircraft is designed to handle disturbances effectively, which is essential for performance.



13 MANUFACTURING PLAN

We focused on finding lightweight, cost-effective materials to avoid production delays and maintain a smooth manufacturing timeline. We carefully balanced weight reduction with the need for structural integrity. After evaluation, we chose thermos-col for the outer covering and balsa wood for internal parts due to their strength-to-weight ratios. Our manufacturing plan included precise cutting, shaping, and assembling with quality control to meet strict design standards. This process allowed us to create a high-performance aircraft within budget, ready for the DBFC competition.

13.1 MATERIAL SELECTION

Our team's experience with balsa, plywood, thermos-col, and depron guided our material selection for this project. We chose depron and thermos-col for the lightweight properties and strength, using balsa for the longerons to ensure the aircraft can handle flight stresses. Our manufacturing plan includes careful cutting, shaping, and assembling with quality control to meet strict design standards. By leveraging our experience, we aim to build a high-performance aircraft that fits our budget while ensuring strong structural performance to meet competition demands.

13.2 PRIMARY MATERIAL

Thermo-col is basically the term for expanded polystyrene, used due to its excellent light-weighting properties and for various applications. Analogously, Depron sheets are often used in an RC airplane. These are often utilized in making the wings since they offer good stiffness and weigh very light. Both are very crucial to bring down overall weight. Lighter means the weight that would enable an RC airplane to better maneuver. It improves the lift-to-weight ratio by making optimal use of Depron in wings and thermos-col in other components. Therefore, flying time can be increased with more acute maneuvers.



Figure 13-1 Depron Sheet



Figure 13-2 Thermo-col

Depron and thermos-col are lightweight yet structurally strong materials ideal for RC planes. Depron helps maintain stability and resilience during maneuvers, while thermos-col absorbs shocks. Together, these materials provide a strong, lightweight structure, offering stability, reliability, and high performance while meeting strict flight requirements.

13.3 SECONDARY MATERIAL

Carbon fiber is a key material for RC planes due to its high strength, rigidity, and excellent strength-to-weight ratio. It reduces the aircraft's weight while improving the lift-to-weight ratio. Carbon fiber can be molded into aerodynamic shapes, enhancing efficiency and reducing drag. Available in sheets, rods, and woven fabrics, it can be selected based on specific needs. Balsa wood is used to reinforce critical areas of the carbon fiber fuselage, adding strength without adding weight.



Figure 13-3 Carbon Fiber Rod



13.4 MANUFACTURING PROCESS SELECTION

13.4.1 Cutting Materials (Sawing, Trimming)

We carefully selected materials and tools for precise, high-quality construction. For balsa wood, we used a fine-toothed hand saw for clean cuts and a utility knife for fine adjustments. Depron wings were shaped with a utility knife and metal ruler to ensure smooth, symmetrical cuts. For the thermos-col fuselage, we used a hot wire cutter to shape the material without causing it to crumble, resulting in a smooth, contoured shell.

13.4.2 Shaping and Smoothing (Sanding, Grinding)

In crafting the aircraft components, we applied specific sanding techniques to ensure smooth finishes and precise fits. For balsa wood, used in longerons, spars, and bulkheads, we used a sanding block/sandpaper after cutting to refine edges and achieve a clean, precise shape. A Dremel tool provided additional precision for fine adjustments. For the depron wings, light sanding was applied to smooth the edges, ensuring symmetry in the wing profile. The thermos-col fuselage required fine sanding to create a smooth, contoured outer shell without crumbling, resulting in a streamlined finish.

13.4.3 Forming the Wing

To build the aircraft wing, we used epoxy or CA glue to attach spars to the depron wing, ensuring a strong bond without adding weight. Precision adjustments were made with a Dremel tool for perfect fit. The wing was covered with heat shrink film after sanding the edges to create a smooth, streamlined surface, enhancing both structural integrity and aerodynamic performance.



Table 13-1 Forming Wing

Type	Name	Used for
Forming the Wing	Attaching Spars to Thermocol Wing	
	Epoxy or CA Glue	We applied glue sparingly, which provided a solid bond without adding weight.
	Dremel Tool	After initial cuts, we used a hobby knife to make fine adjustments for an exact fit.
	Covering the Wing	
	Heat Shrink Film	We used light sanding to make our wing cuts more smooth resulting in a good streamline flow

13.4.4 Fuselage Assembly

For the fuselage assembly, we used specific adhesives and alignment techniques to ensure a sturdy structure. Epoxy was applied to larger joints for a strong bond, while foam-safe CA glue was used on smaller areas to prevent damage to the thermos-col shell and balsa bulkheads. For bulkhead alignment, T-pins and glue were used to secure the bulkheads, maintaining the fuselage's shape and integrity during construction.

Table 13-2 Fuselage Assembly

Type	Name	Used for
Fuselage Assembly	Thermo-col Shell and Balsa Bulkheads	
	Epoxy or CA Glue	Epoxy was used for larger joints, and foam-safe CA glue for smaller areas to avoid damaging the thermos-col.
	Bulkhead Alignment	
	Pinning and Gluing	Bulkheads were secured with T-pins and glue, ensuring the fuselage maintained its shape



13.4.5 Final Steps and Testing

During the final testing stage, we focused on refining the aircraft's structure for optimal aerodynamic performance. We used fine-grit sandpaper to smooth joints and edges, reducing drag and improving airflow over the surface. To protect and finish the aircraft, acrylic spray paint was applied, giving it a polished appearance. Additionally, bulkheads were secured with T-pins and glue to ensure that the fuselage maintained its shape, contributing to the overall stability and durability of the design.

Table 13-3 Final Assembly

Type	Name	Used for
Final Testing	Final Smoothing and Sanding	
	Epoxy or CA Glue	We used fine-grit sandpaper to smooth joints and edges, enhancing aerodynamics by reducing drag.
	Acrylic Spray Paint	
	Pinning and Gluing	Bulkheads were secured with T-pins and glue, ensuring the fuselage maintained its shape

13.4.6 Manufacturing Process Investigated

During our project, we conducted detailed research on various manufacturing techniques that would ensure that our RC plane was constructed successfully using thermocol, balsa wood, and depron. This section reviews the techniques applied and offers what we have learned.



13.5 MATERIAL SELECTION

For the fuselage, there was thermocol material with lightweight and easy-to-form properties. Balsa wood has been chosen for bulkheads and longerons because of more strength. The wings were designed with depron sheets to maintain an ideal balance between weight and aerodynamics. These materials made the design light yet robust.

13.5.1 Construction Techniques

The predominant construction techniques implemented were accurate cutting and trimming of the thermos-col as well as balsa wood. Fuselage and wing components were assembled using manual tools along with adhesives. It was an extremely important addition to the fuselage as carbon fiber rod, which gives strength without excessive weight gain. We saw that only proper alignment during assembling could hold the aerodynamics intact.

13.5.2 Key challenges

Some of the key challenges included achieving the weight distribution that would enable flight at the optimal performance level. The solution came about in the form of iteration with component placement change until balance was achieved. Secondly, the materials were soft and needed gentle handling when putting them together not to break.

13.5.3 Lessons learned

Lessons learned in the fabrication process included the critical importance of proper planning and testing throughout the manufacturing process. Proto-type runs permitted us to discover weaknesses in the original design and adjust accordingly. For example, it was learned that adhesiveness to joints through additional applied adhesives improved structure durability.



13.5.4 Future Recommendations

In future designs, using tougher foam with higher elasticity and advanced manufacturing processes like CNC precision cuts could improve the RC plane's performance and quality. The materials we chose had a big impact on the plane's ability to fly longer and maneuver better. By carefully selecting and combining the right materials, we were able to create an effective design that met our goals for the competition

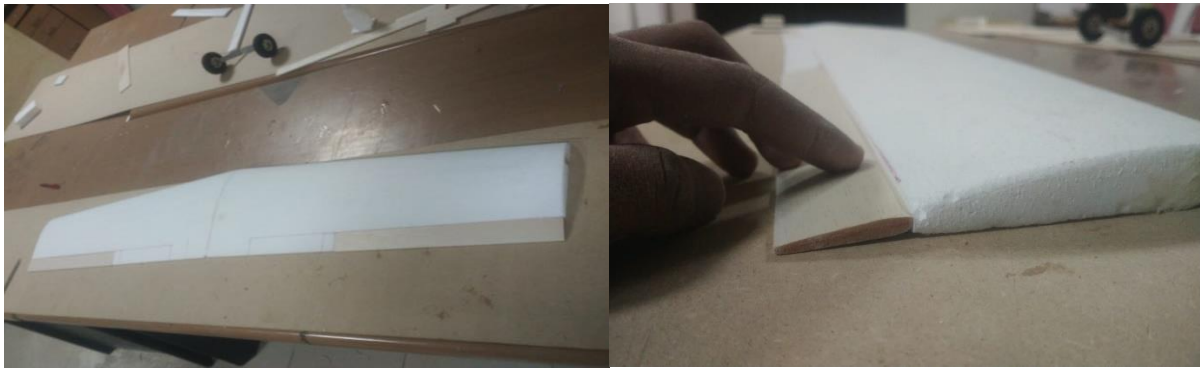


Figure 13-4 Fabrication of Wing



Figure 13-5 Fabrication of Fuselage

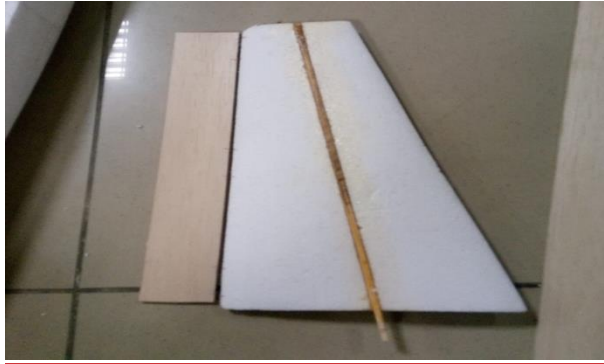


Figure 13-6 Empennage Fabrication



Figure 13-7 External Structure of Aircraft



13.5.5 Conclusion

Thus, we should conclude that it was through careful material choice and rigorous adherence to the manufacturing process that helped in the successful construction of our RC plane. The use of thermocol for the fuselage, balsa wood for the longerons and the bulkheads, and carbon fiber rod for the spar still made the model lightweight but structurally rigid. An aerodynamic efficiency made the plane because the depron from which the wing was made.

Precision was the word in every step involved in making the RC plane. Paying attention to minute details led to optimizing its performance and durability. A combination of materials chosen and hands-on experience combined with teamwork has enabled us to come up with this innovative RC plane, keeping it balanced in terms of performance, weight, and cost-effectiveness. It's a project that has portrayed our engineering capability and even emphasized the selection of the right material with thoughtfulness along with collaborative effort to make the design come into effect.



14 TESTING PLAN

14.1 Objectives of Testing

The primary objectives of this testing plan are to validate the functionality and robustness of each UAV subsystem, confirm mission readiness, and assess overall performance capabilities. Each subsystem, including the motor, avionics, battery, and control systems, will be rigorously tested to ensure operational reliability. Additionally, the UAV must demonstrate full readiness for mission conditions, encompassing specific maneuverability, payload handling, and endurance requirements as outlined by the competition. By meeting these objectives, we will confirm that the UAV can consistently execute the required maneuvers, transport payloads effectively, and sustain flight for the necessary duration, ensuring mission success.

14.2 SUBSYSTEM TESTING

14.2.1 Motor and Propulsion

The motor and propulsion testing are designed to confirm the motor's ability to provide sufficient thrust and stable performance over extended operation. During these tests, the propulsion system will operate under simulated flight loads to measure maximum thrust, response times, and energy consumption. Key indicators such as RPM consistency, motor temperature, and power efficiency will be recorded across different power levels, ensuring that thrust output remains stable and temperatures are within safe operating limits.



Figure 14-1 Thrust testing stand made by teams



Figure 14-2 Propulsion system testing

14.2.2 Avionics and Control System

The avionics and control system testing aims to validate smooth communication and precise response between the flight controller, servos, ESCs, and small control unit. Pre-flight checks will ensure servo alignment, signal strength, and control accuracy. Fail-safe mechanisms will also be tested to confirm that recovery procedures are activated as needed. Successful tests will be indicated by rapid, accurate servo movements with no signal drops, as well as reliable fail-safe engagement during simulated loss of control.



14.2.3 Battery and Power System

Battery and power system testing will focus on discharge rates, voltage stability, and endurance under mission-specific power demands. By placing the battery under the expected load conditions, we will monitor voltage consistency and heat levels throughout the flight duration. The battery will be evaluated for stable power delivery without voltage drops, sufficient operational time for mission completion, and a controlled temperature profile during discharge.

3. STRUCTURAL INTEGRITY AND LOAD TESTING

Wing load testing will confirm the wings' capacity to withstand aerodynamic forces encountered during flight. This will involve incrementally increasing the load applied to the wings, observing for any deformation under the maximum expected forces. Repetitive load applications will simulate the stress experienced in flight. The wings must tolerate these loads without permanent deformation and show minimal deflection under pressure, ensuring structural resilience.

Motor mount and fuselage testing will verify the UAV's stability when exposed to propulsion vibrations and dynamic forces. Through force and vibration tests, the motor mount and fuselage will be observed for any signs of wear, loosening, or cracks. The evaluation criteria will include maintaining motor alignment and structural integrity under high-vibration conditions, with no visible damage or instability during testing. This will ensure the UAV's body and propulsion mounts remain secure and perform reliably under flight stresses.



14.3 CHECKLISTS

Table 14-1 Propulsion Checklist

General	Propulsion test	Aerodynamics test	Final inspection
Spotter ready	Check battery voltage	Wing secured	Roads clear
Sensors calibrated	Connection secured	Flaps fixed	Driver ready
Receiver ready	Propeller secured	Ground clearance	Operator ready
Servos working	Transmitter check	AoA fixed	

14.3.1 Preflight check list

Table 14-2 Pre Flight-Checklist

Overview	Before power up	After power up	Final inspection
Date & time	Check battery voltage	Control surface check	Pilot and spotter check
Wind speed/direction	Connection secured	Fail safe	Clear road
location	Check cg	Transmitter check	
Aircraft			
Battery			
Additional info			

14.4 FLIGHT TEST SCHEDULE AND MISSION SIMULATIONS

The pre-flight test phase ensures all UAV systems are working properly before mission tests. It includes short flights to check take-off, landing, control, and maneuverability. For Mission 1, the UAV's handling and stability without a payload will be tested through maneuvers like rolls and loops. In Mission 2, the UAV's stability with a payload will be assessed, focusing on



payload security and maneuver execution. Mission 3 will test endurance, with the UAV carrying a payload for a long flight and performing accurate payload drops. Successful completion of these tests ensures the UAV is ready for all mission requirements.

14.5 Post-Flight Analysis

Upon completion of each test, all recorded data on subsystem performance, structural integrity, and flight stability will be compiled for detailed analysis. Any deviations or unexpected outcomes will be thoroughly reviewed to identify potential areas for refinement. Based on these insights, adjustments will be made to the UAV's components or settings, optimizing its performance for future tests. This final stage of analysis and adjustment ensures that the UAV is operating at peak performance and is fully prepared for the competition.



15 REFERENCES

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16 APPENDIX

16.1 Aircraft Design Calculations Code

```
clc
clear all
%% Sizing of Aircraft:

rho= 1.155; %% Density of Air in Kg/m^3
V= 25;      %% Velocity in m/s
W = 2.2    %%Weight of Aircraft in kg
Wing_Loading = 7 %% Select a suitable Wing loading using
Raymer

S_W = W/Wing_Loading; %% Surface Area of Wing

%% For Calculating Cl of airfoil:

q= 1/2*rho*(V*V); %% Dynamic Pressure

Cl_aircraft = W/(q*S_W)
Cl_wing = Cl_aircraft/0.9
Cl_airfoil= Cl_wing/0.85;

%% Geometric Parameters of Wing

lambda_Wing= 0.75;
AR= 6.5; %% 6:1 range is easier to build and more likely to
be successful for RC Plane

X= (1+lambda_Wing+(lambda_Wing^2))/(1+lambda_Wing);
b= sqrt(AR*S_W);

C_root_Wing = (2*S_W)/(b*(1+lambda_Wing));
C_tip_Wing = C_root_Wing*lambda_Wing;
MAC_Wing = (2/3)*C_root_Wing*X ;%% Mean Aerodynamic Chord
of Wing

%% Geometric Parameters of Faueslage
```




```
F_Length= 0.75*b;

%% Empennage Sizing

C_Vh= 0.425; %% For RC Plane we have a range of 0.35-0.5
C_Vv= 0.03;  %% For RC Plane we have a range of 0.02-0.04

LH= 0.7*F_Length; %% LH is 70% of Fuselage Length
LV=LH;

%% For Horizontal Stabilizer

AR_HT= 4; %% For other planes (i.e RC planes) we have a
range of 3-5;
lambda_HT= 0.45; %% For other planes (i.e RC planes) we
have a range of 0.3-0.6;

S_HT= (C_Vh*S_W*MAC_Wing)/LH;
X_HT= (1+lambda_HT+(lambda_HT^2))/(1+lambda_HT);
b_HT= sqrt(AR_HT*S_HT);
chord_Ht= b_HT/AR_HT;
C_root_HT = (2*S_HT)/(b_HT*(1+lambda_HT));
C_tip_HT = C_root_HT*lambda_HT;
MAC_HT = (2/3)*C_root_HT*X_HT; %% Mean Aerodynamic Chord
of Horizontal Tail

%% For Vertical Stabilizer

AR_VT= 1.65; %% For other planes (i.e RC planes) we have a
range of 1.3-2;
lambda_VT= 0.45; %% For other planes (i.e RC planes) we
have a range of 0.3-0.6;

S_VT= (C_Vv*S_W*b)/LV;
X_VT= (1+lambda_VT+(lambda_VT^2))/(1+lambda_VT);
b_VT= sqrt(AR_VT*S_VT);
chord_Vt= b_VT/AR_VT;
C_root_VT = (2*S_VT)/(b_VT*(1+lambda_VT));
C_tip_VT = C_root_VT*lambda_VT;
MAC_VT = (2/3)*C_root_VT*X_VT; %% Mean Aerodynamic Chord
of Horizontal Tail
```



```
fprintf("The Calculated Data is as below ; \n\n")

fprintf("Surface Area of Wing(S) = %f \n ",S_W);
fprintf("Coefficient of Lift of airfoil (Cl_airfoil) = %f
\n ",Cl_airfoil);
fprintf("Wing Span is of Wing (b) = %f \n ",b);
fprintf("Root Chord of Wing (C_root) = %f \n
",C_root_Wing);
fprintf("Tip Chord of Wing (C_tip) = %f \n ",C_tip_Wing);
fprintf("Mean Aerodynamic Chord of Wing (c_bar) = %f \n\n
",MAC_Wing);

fprintf("Geometric Parameters of Fuselage:\n\n")
fprintf("Length of Fuselage = %f\n\n",F_Length);

fprintf("Horizontal Stabilizer:\n\n")

fprintf("Moment Arm of Horizontal Tail (LH) = %f \n ",LH);
fprintf("Surface Area of Horizontal Tail = %f \n ",S_HT);
fprintf("Wing Span is of Horizontal Tail(b_HT) = %f \n
",b_HT);
fprintf("Chord of Horizontal Tail (C) = %f \n ",chord_Ht);
fprintf("Root Chord of Horizontal Tail (C_root) = %f \n
",C_root_HT);
fprintf("Tip Chord of Horizontal Tail (C_tip) = %f \n
",C_tip_HT);
fprintf("Mean Aerodynamic Chord of Horizontal Tail (c_bar)
= %f \n\n ",MAC_HT);

fprintf("Vertical Stabilizer:\n\n")

fprintf("Moment Arm of Vertical Tail (LH) = %f \n ",LV);
fprintf("Surface Area of Horizontal Tail = %f \n ",S_VT);
fprintf("Wing Span is of Vertical Tail (b_vt) = %f \n
",b_VT);
fprintf("Chord of Horizontal Tail (C) = %f \n ",chord_Vt);
fprintf("Root Chord of Vertical Tail (C_root) = %f \n
",C_root_VT);
fprintf("Tip Chord of Vertical Tail (C_tip) = %f \n
",C_tip_VT);
```



```
fprintf("Mean Aerodynamic Chord of Vertical Tail (c_bar) =  
%f \n ",MAC_VT);
```